Market Risk Modelling

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Frankfurt School of Finance & Management
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Summer 2026

Preface

This course covers fundamental approaches and techniques of financial risk management. For the example of market risk, we investigate risk factors, risk measures and various methods of quantifying market risks, such as

- descriptive statistics of risk factors,
- dimension-reduction techniques (factor models)
- estimation techniques (maximum likelihood),
- modelling of financial time series (in particular, GARCH models),
- dependence modelling (copulas).

Particular emphasis is put on practical applications in group projects.

Contents

Self-study

- This module is supposed to provide approximately 150 hours of course work consisting of lectures, a group project, and independent study.
- This means that you are expected to invest a substantial amount of time for preparation of my lectures and in addition spend a substantial amount of self-study time on parts of my lecture notes, which I do not cover in detail during my lectures.
- You are expected and invited to raise questions on parts of the lecture material that we do not cover thoroughly during my lectures.
- Your grades can depend on your participation in my lectures, as outlined below.

Group projects: Grading I

I will assign individual grades, which depend on:

- the group's grade for the project;
- your individual contribution to the group project;
- potentially also your contributions during my lectures.

I decided to take this approach because many groups in the past complained that individual students attempted to free-ride or at least that individual contributions to the group's project were very unequal. The grading approach outlined below incentivizes each student to work hard on the project, as the quality and quantity of individual contributions will be made transparent and can measurably impact the individual grade. This approach is acceptable for Frankfurt School's Examination Board.

I will assess your individual contributions to your group project as follows:

- I will first and foremost rely on my own impressions that I gain during my meetings with the project teams.

Group projects: Grading II

- To challenge my own impressions of your individual contributions to your projects, I may consider the results of a peer assessment: I expect from every team member after submission of the project results an email (**which I will treat as strictly confidential**) in which he or she suggests for every other team member an increase or decrease of the final grade in percentage points, relative to the grade for the project, **under the side condition that the proposed increase or decrease on average equals 0**.
  - Example: All other team members may be of the opinion that one group member contributed a lot more to the project than the rest of the team. These students acknowledge this by stating in their emails on average that this individual's grade should be 10 percentage points higher than the grade for the project. Let us assume that the grade for the project is 75%. I would consequently **consider** to assign an 85%-grade to this individual student, if this grade is in line with my own impressions. Again, the results of the peer assessment do not translate 1:1 into

Group projects: Grading III

individual grades, but I only use the results of the peer assessment to challenge my own impressions.

Group projects: Grading IV

The grade for the project depends on the following:

1. Scope, complexity and correctness of the project submission
   - How realistic are your assumptions?
   - Are you aware of all assumptions made?
   - Do you validate your assumptions properly, e.g. by means of suitable statistical tests?
   - Should you work with portfolio data: How complex is the structure of your portfolio?
   - How much state-of-the-art and how innovative is your solution?
   - Is your solution correct? I will check your codes :-)
   - Do you interpret your results properly?
2. Degree of independent work without significant assistance from the professor

Group projects: Grading V

- I will naturally provide guidance and support to all groups and I am more than happy to assist, should you encounter conceptional, mathematical or coding issues. But please understand that I need to take into account, how independently you worked and how much support you needed. It would be unfair to give the same grade to two groups with roughly similar project results when one group achieved these results through independent work while the other group required considerable assistance and various extra appointments with me.

3. Presentation
   - Did the organisation of the slide deck support the explanation of difficult concepts?
   - Was there enough time for important concepts?
   - Were the ideas presented in an obviously logical (or coherent) sequence?
   - Was the specific purpose of the presentation made clear?
   - Was there a clear introduction & conclusion?
   - Was the presentation insightful?

Group projects: Grading VI

- Were difficult concepts explained sufficiently well?
- Were appropriate examples used?
- Did the audience benefit from the analysis of the topic or of its consequences?
- Did the audience understand something about the topic better because of the presentation?
- Was the methodology explained?
- Was there a clear answer to the main questions?
- Was the overall argumentation convincing?
- **How well could you answer my questions (and potential questions of the audience) after the presentation?**

Group projects: Guidance

- Appropriately validate any assumption made in all above steps.
  - This validation activity may include (but does not need to be restricted to) conducting suitable statistical tests, graphical analyses and impact assessments (how material is this assumption in terms of VaR contribution?).
- Clearly structure and comment programming code.

Contents

Risk factors

Consider a financial position V. This can be a single security or a portfolio consisting of different individual positions such as stocks, bonds, loans and derivatives. For example, V can be the whole portfolio of all contracts of a financial institution or the portfolio (book) of a trading unit.
The value V_0 of the position today is known (may be a bold assumption; think of beginning of financial crisis when there were no market prices for some products and institutions had to resort to mark-to-model!). We are interested in the **change in value** ΔV, **profit and loss, P&L**, over a **time horizon** $[0, \Delta t]$:

$$\Delta V = V_{\Delta t} - V_0.$$

ΔV is a random variable that describes the risk of the position.

Risk factors

To analyze ΔV we write the value V_t of the position at time t as a **function of** d **random risk factors** $\mathbf{Y}_t = (Y_t^1, \dots, Y_t^d)^T$:

$$V_t = f(t, \mathbf{Y}_t). \tag{1}$$

Note that $\mathbf{Y}_t$ denotes a complete **vector of risk factors**. The risk factors $\mathbf{Y}_t$ are assumed to be observable at time t. Examples of risk factors are prices or yields of securities, interest rates, exchange rates and volatilities.
For a given position, the process of identifying the set of driving risk factors and the appropriate function f is called **risk mapping**. The change in value of the position V can then be represented as

$$\begin{aligned}\Delta V &= f(\Delta t, \mathbf{Y}_{\Delta t}) - f(0, \mathbf{Y}_0) \\ &= f(\Delta t, \mathbf{Y}_0 + \Delta\mathbf{Y}) - f(0, \mathbf{Y}_0)\end{aligned} \tag{2}$$

with **factor changes**

$$\Delta \mathbf{Y} = \mathbf{Y}_{\Delta t} - \mathbf{Y}_0$$

over the time horizon $[0, \Delta t]$. Applying Taylor's formula

$$f(x) = f(x_0) + f'(x_0)\Delta x + \frac{f''(x_0)}{2}(\Delta x)^2 + \cdots$$

we get a **first order approximation** for the change in value

$$\Delta V \approx \frac{\partial}{\partial t} f(0, \mathbf{Y}_0)\Delta t + \sum_{j=1}^{d} \frac{\partial}{\partial y_j} f(0, \mathbf{Y}_0) \cdot \Delta Y^j. \tag{3}$$

We shall see later that this approximation is very useful and frequently used in practice.

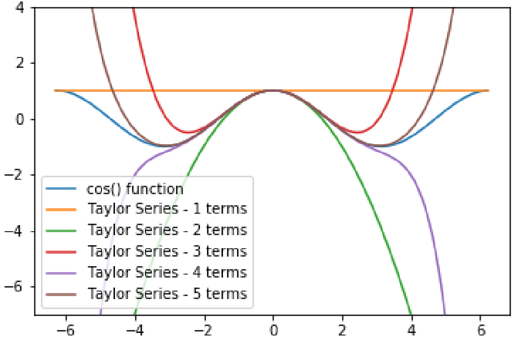

Figure: Illustration of the Taylor formula

Risk factors

One can also express the change of a risk factor by its **relative change (percentage change)** (risk factor return) $R(\mathbf{Y})_{\Delta t} = (\Delta Y^1/Y_0^1, \dots, \Delta Y^d/Y_0^d)^T$

$$\Delta \mathbf{Y} = \mathbf{Y}_{\Delta t} - \mathbf{Y}_0 = \mathbf{Y}_0 \cdot R(\mathbf{Y})_{\Delta t}, \tag{4}$$

where the multiplication of the vectors $\mathbf{Y}_0$ and $R(\mathbf{Y})_{\Delta t}$ is component-wise. The advantage of using percentage changes is that their probability distribution turns out to be more stable over time. This is extremely important for modelling future changes by models whose parameters are estimated from historical data.

The above first order approximation of the P& L, Equation (3), can now be written using the risk factor returns,

$$\Delta V \approx \frac{\partial}{\partial t} f(0, \mathbf{Y}_0)\Delta t + \sum_{j=1}^{d} Y_0^j \frac{\partial}{\partial y_j} f(0, \mathbf{Y}_0) \cdot R(Y^j)_{\Delta t}, \tag{5}$$

with so-called **delta equivalents**

$$Y_0^j \frac{\partial}{\partial y_j} f(0, \mathbf{Y}_0). \tag{6}$$

The delta equivalent allows for an interpretation as the factor translating percentage changes of the corresponding risk factor into implied value changes of the underlying position.

Example 1. Consider a portfolio V consisting of the securities $S^1, \dots, S^n$

$$V_t = \sum_{i=1}^{n} a_i S_t^i.$$

Obviously, the driving risk factors are the securities $S^1, \dots, S^n$ themselves.

Example 2.
A EUR currency based investor holds a position $a\,S_t$ in a USD denominated stock S_t. Denote by X_t the foreign exchange rate, i.e., the time-t value of one USD expressed in EUR. For the investor the value of his position is then

$$V_t = a\,S_t\,X_t.$$

Value changes, P&L, are now driven by two risk factors.

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Example 3.
Consider a position consisting of a default risky bond or loan B. Often the price of a bond is expressed with respect to a benchmark zero rate curve $R_t^{T1}, \ldots, R_t^{TN}$ (government curve, swap curve) and a spread S quantifying the default risk of the issuer:

$$B_t = B(t, R_t^{T1}, \ldots, R_t^{TN}, S_t) = \sum_{i:T_i>t} C_i \exp\left(-(R_{[t,T_i]} + S_t)(T_i - t)\right). \qquad (7)$$

Thus, the bond's value is a function of zero interest rates and the credit spread.

Contents

Example for risk factor identification: Principal Component Analysis

- **Principal Component Analysis (PCA)** is a standard method for reducing the dimension of high dimensional, highly correlated systems.
- The common factors are unobservable (latent) and are directly estimated from the data.
- While being explanatory in a statistical sense, the factors may not have an economic interpretation.

Example for risk factor identification: Principal Component Analysis

- The objective is to take p random variables $X_1, X_2, \dots, X_p$ and find (linear) combinations of these to produce random variables $Z_1, \dots, Z_p$, the **principal components**, that are uncorrelated.
- Geometrically, expressing $X_1, \dots, X_p$ through linear combinations $Z_1, \dots, Z_p$ can be thought of as shifting and rotating the axes of the coordinate system (see left graph of the Figure on page 27).
- Hence, the transform leaves the data points unchanged, but expresses them using different coordinates.
- The lack of correlation is a useful property because it means that the principal components are measuring different "dimensions" of the data.
- The principal components can be ordered according to their variance, that is, $\mathrm{Var}(Z_1) \geq \mathrm{Var}(Z_2) \geq \cdots \geq \mathrm{Var}(Z_p)$.
- If the variance captured in the higher dimensions is sufficiently small, then discarding those higher dimensions will retain most of the variability, so only little information is lost.

Example for risk factor identification: Principal Component Analysis

Example:

- Let (X, Y) be are standard normally distributed random variables with a correlation of 0.7.
- The left graph of the Figure on page 27 shows a scatterplot of a sample of 100 random numbers $(x_1, y_1), \ldots, (x_{100}, y_{100})$.
- By shifting and rotating the axes, new variables Z_1, Z_2, the principal components, with $Z_i = a_i X + b_i Y$ are obtained.
- The data sample expressed in terms of Z_1, Z_2 is uncorrelated.
- Also, the variance of the data contribution from the first principal component Z_1 is much greater than the variance contribution from the second principal component.
- The graph on the right shows the data points when the data are onto the first principal component, discarding the second dimensions.

Example for risk factor identification: Principal Component Analysis

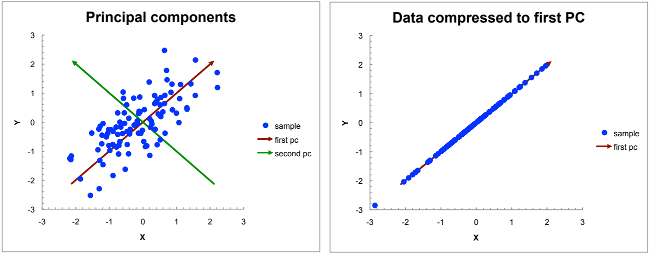


Figure: PCA example.

Example for risk factor identification: Principal Component Analysis

- The sample variance of $(x_1, \dots, x_n)$ is 0.9244 and the sample variance of $(y_1, \dots, y_n)$ is 0.9226, whereas the sample variance of the data in the first principal component is 1.6014 and of the second principal component is 0.2456.
- In other words, while the original axes each account for roughly 50% of the total variance, the first principal component accounts for 87% of the sample variance.
- Neglecting the second principal component and expressing the data in the first principal component only retains 87% of the variance (see right graph of Figure 2).
- In practice, the number of dimensions will be higher, and one will choose the number of principal components to reflect a certain variance contribution such as 99%.
- If the data are sufficiently correlated, then only few dimensions (principal components) will be required even for high-dimensional data.

Example for risk factor identification: Principal Component Analysis

- Most statistics or numerical software packages offer the functionality to calculate the principal components (the **eigenvalues** in general terminology from Linear Algebra).

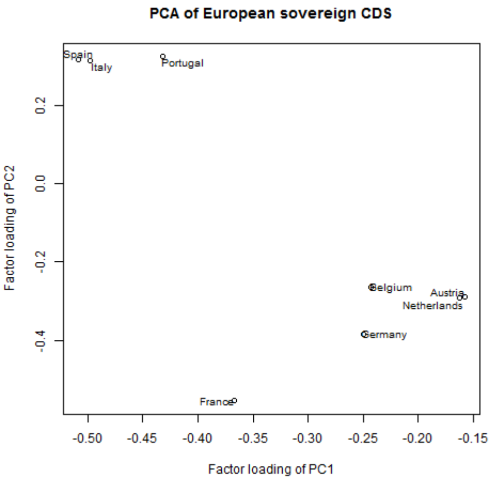


Figure: PCA of CDS returns - Practical application.

Contents

Denote by S_t the **price** of a **security** S at time t. Today, i.e., at $t = 0$, the price S_0 is known. For $t > 0$, the value S_t is a random variable. The **discrete return** $R^d_{[t,t+\Delta]}$ and the **continuous return** $R^c_{[t,t+\Delta]}$ over the time horizon $[t, t + \Delta]$ are defined by

$$R^d_{[t,t+\Delta]} = \frac{S_{t+\Delta} - S_t}{S_t} \tag{8}$$

and

$$R^c_{[t,t+\Delta]} = \ln \frac{S_{t+\Delta}}{S_t} = \ln S_{t+\Delta} - \ln S_t \tag{9}$$

We thus ignore intermediate cash flows from the security such as, for example, dividends, or we assume that such cash flows are reinvested in the security and therefore reflected in the price. Then, $S_{t+\Delta} = S_t(1 + R^d_{[t,t+\Delta]})$ and $S_{t+\Delta} = S_t \exp(R^c_{[t,t+\Delta]})$, respectively.

Returns

From the point of view of time t, returns are random variables and known only at the end $t + \Delta$ of the reference period. We write R_t^d resp. R_t^c for the discrete resp. continuous return over the period $[0, t]$. The price S_t of the security at time t can then be expressed as a function $S_t(R)$ of its return:

$$S_t = S_t(R_t^d) = S_0(1 + R_t^d), \quad S_t = S_t(R_t^c) = S_0 \exp(R_t^c).$$

Given a portfolio P of securities $S^1, \dots, S^n$ with number of shares $a_1, \dots, a_n$,

$$P = \sum_{i=1}^{n} a_i S^i,$$

the discrete return of the portfolio turns out to be just the weighted sum of the discrete returns $R_t^{d,i}$ of the securities S^i

$$R_t^{d,P} = \sum_{i=1}^{n} w_i R_t^{d,i}, \tag{10}$$

with weights

$$w_i = \frac{a_i S_0^i}{\sum_{i=1}^{n} a_i S_0^i}.$$

Returns: Example

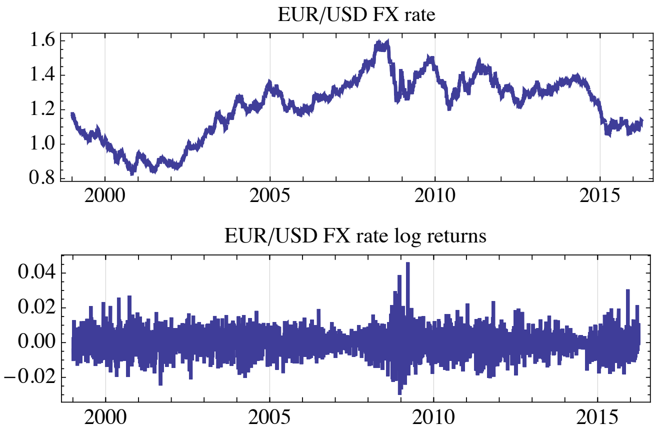


Figure: Daily prices of EUR/USD exchange rate and corresponding log-returns. Data source: Federal Reserve

Log-returns vs. discrete returns

- When do we use discrete returns and when do we use log-returns?
- If the time scale Δt is small (e.g. one day or one week), then the differences between the two types ore returns are negligible (see next slide).
- Hence, it depends on the application:
  - Discrete returns are additive across portfolios:

$$r_p^d = \sum_{i=1}^{n} w_i \, r_i^d ,$$

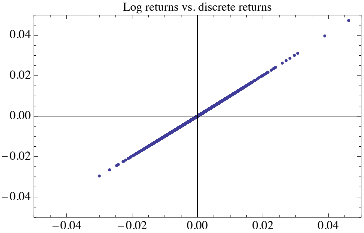
    where $(w_1, \dots, w_n)$ are the (percentage) weights of the portfolio constituents.
  - Log-returns are additive across time:

$$r_{t,t+2}^c = \ln\left(\frac{S_{t+2}}{S_t}\right) = \ln\left(\frac{S_{t+2}}{S_{t+1}} \frac{S_{t+1}}{S_t}\right) = \ln\left(\frac{S_{t+2}}{S_{t+1}}\right) + \ln\left(\frac{S_{t+1}}{S_t}\right) .$$

Log-returns vs. discrete returns

Here are some ways to measure the similarity of log vs. discrete returns:

1. A **scatter plot** shows the degree of similarity, e.g. for the EUR/USD return data:



2. The degree of the linear relationship is measured by the correlation coefficient, e.g. 0.999979 for EUR/USD returns.
3. A first-order Taylor approximation of the log-return gives the corresponding discrete return:

$$\ln\left(\frac{S_{t+1}}{S_t}\right) = \ln(S_{t+1}) - \ln(S_t) = \Delta \ln S \approx \frac{1}{S_t}\Delta S.$$

The approximation is valid under the assumption of small ΔS, which is supported by small Δt.

Contents

Risk measures

Risk measures characterize possible changes in the value of a position V. State-of-the-art risk measures characterize the probability distribution of the possible value changes ΔV. They are named **distribution-based risk measures**. In this framework risks are aggregated over sub-positions taking into account netting and diversification effects automatically.

The crucial input is of course the specification of the probability distribution of the value changes aggregated over risk factors. In this context we face two critical problems:

1. The probability distributions of the driving factors, such as the parameters of the distribution, have to be estimated from historical data. It is questionable whether information derived from past data is still valid for modelling the future. For example there can be structural changes.
2. Taking into account all possible risk factors for a big portfolio can be too demanding and is impossible to realize with reasonable effort. One is often forced to approximate the distribution of the overall value changes. This entails making several simplifying assumptions.

The classical distribution-based risk measure is the **variance** $\mathrm{Var}(\Delta V)$ **of the value change** ΔV. Markowitz' portfolio theory is one of the origins of variance as risk measure. Since variance measures the squared dispersion from the mean in both directions it is a reasonable risk measure only in case of a symmetric distribution such as, for example, the normal distribution.

Contents

Today's most popular distribution-based risk measure is **Value at Risk**.

Definition 5.1

For a position V with random changes in value ΔV over a time horizon $[0, \Delta t]$ the Value at Risk at the confidence level $\alpha \in (0,1)$ is given by the smallest number l such that the probability that the loss $-\Delta V$ exceeds l is no larger than $(1-\alpha)$. Formally,

$$\begin{aligned}\mathbf{VaR}_{\alpha,\Delta t} &= \inf\{l \in \mathbb{R} : \mathbf{P}(-\Delta V > l) \leq 1-\alpha\} \\ &= \inf\{l \in \mathbb{R} : \mathbf{P}(-\Delta V \leq l) \geq \alpha\} = Q_\alpha(-\Delta V).\end{aligned}$$

Alternatively (observe that a discrete distribution function is right-continuous)

$$\mathbf{P}(-\Delta V > \mathbf{VaR}_\alpha) \leq 1 - \alpha < \mathbf{P}(-\Delta V > x), \quad \text{for all } x < \mathbf{VaR}_\alpha,$$

or, equivalently,

$$\mathbf{P}(-\Delta V \leq x) < \alpha \leq \mathbf{P}(-\Delta V \leq \mathbf{VaR}_\alpha), \quad \text{for all } x < \mathbf{VaR}_\alpha.$$

In case the distribution density of the value changes is everywhere strictly positive (i.e., the cumulative distribution function is strictly increasing), **VaR** is characterized by the condition

$$\mathbf{P}(\Delta V \leq -\mathbf{VaR}_{\alpha,\Delta t}) = 1 - \alpha, \tag{11}$$

and we have the relationship

$$\mathbf{VaR}_{\alpha,\Delta t} = Q_{\alpha}(-\Delta V) = -Q_{1-\alpha}(\Delta V). \tag{12}$$

One reason for the popularity of **VaR** as risk measure is that it is easy to interpret and can be applied consistently over all categories of risk. However, a major **disadvantage** is that the quantity **VaR** does not differentiate between the severity of losses below $-\mathbf{VaR}$. Moreover, it turns out that in particular situations **VaR** can lead to an unreasonable aggregation of risks over sub-positions.

On the history of VaR: VaR was developed by JP Morgan in the early 1990s.
Example for problem with not differentiating below VaR:

- VaR does not convey information about the risk below its confidence level. If there is a VaR-event you have no idea how bad it is going to be. A common misinterpretation of VaR is to suppose that VaR gives you the actual loss at a certain confidence level.
- This can be exploited e.g. by traders with asymmetric positions that generate a steady small return and only very rarely losses, which are then huge. VaR ignores the downside risk if it falls below a certain confidence level (e.g. CDS).

Some more from a NY time article:

- "VaR may have been a flawed number, but it was the best number anyone had come up with." (Rick Bookstaber)
- VaR is even good as an indicator when numbers seem "off", or when it starts to "miss" on a regular basis. It either means that there is something wrong with the way it's calculated or that the market is no longer behaving "normally".
- VaR changes are important - rather than the actual number, it is important to see where risk is going on a day-to-day basis. "That is probably another danger: because we put a dollar number to it, they attach a meaning to it."
- "That $50 million (VaR) wasn't just the most you could lose 99 percent of the time. It was the least you could lose 1 percent of the time."
- "We had a trade in the mortgage universe where the VaR was $1 million. The same trade a week later had a VaR of $6 million. If you tell me my risk hasn't changed – I say yes it has!" (David Viniar of Goldman Sachs)

Contents

Another reasonable risk measure is the so-called **Expected Shortfall**, which additionally takes into account the severity of losses below $-\mathbf{VaR}$. We give its definition in case the distribution of ΔV admits a density $f_{\Delta V}$.

Definition 5.2

For a position V with random change in value ΔV over the time horizon $[0, \Delta t]$ **Expected Shortfall** $\mathbf{ES}_{\alpha,\Delta t}$ for the confidence level $\alpha \in (0,1)$ is defined as the conditional expected value of all losses exceeding $\mathbf{VaR}_{\alpha,\Delta t}$,

$$\begin{aligned}
\mathbf{ES}_{\alpha,\Delta t} &= \mathbb{E}(-\Delta V \mid -\Delta V \geq \mathbf{VaR}_{\alpha,\Delta t}) \\
&= -\mathbb{E}(\Delta V \mid \Delta V \leq -\mathbf{VaR}_{\alpha,\Delta t}) \\
&= -\frac{\int_{-\infty}^{-\mathbf{VaR}_{\alpha,\Delta t}} x f_{\Delta V}(x) \mathrm{d}x}{1-\alpha}.
\end{aligned}$$

Intermediate steps used for Definition 4.2

$$
\begin{aligned}
&ES_{\alpha,\Delta t}(-\Delta V)\\
&= \mathbb{E}\left(-\Delta V \,|\, -\Delta V \geq VaR_{\alpha,\Delta t}\right)\\
&= \frac{\mathbb{E}\left(-\Delta V \cap -\Delta V \geq VaR_{\alpha,\Delta t}\right)}{\mathbb{P}\left(-\Delta V \geq VaR_{\alpha,\Delta T}\right)}\\
&= \frac{\mathbb{E}\left(-\Delta V \cap -\Delta V \geq VaR_{\alpha,\Delta t}\right)}{1-\alpha}\\
&= \frac{\mathbb{E}\left(-\Delta V \cdot \mathbf{1}_{\left\{-\Delta V \geq VaR_{\alpha,\Delta t}\right\}}\right)}{1-\alpha}\\
&= -\frac{\mathbb{E}\left(\Delta V \cdot \mathbf{1}_{\left\{\Delta V \leq -VaR_{\alpha,\Delta t}\right\}}\right)}{1-\alpha}\\
&= -\frac{\int_{-\infty}^{-VaR_{\alpha,\Delta t}} x \cdot f_{\Delta V}(x)dx}{1-\alpha}
\end{aligned}
$$

Contents

ES estimator 1

Let $m := (n+1)(1-\alpha)$. Then,

$$\widehat{ES}_{\alpha}^{(1)}(X) = -\frac{1}{\lfloor m \rfloor} \sum_{i=1}^{\lfloor m \rfloor} X_{(i)}$$

ES estimator 2

$$\widehat{ES}_\alpha^{(2)}(X)$$

$$= -\frac{1}{\lfloor m \rfloor + 1}\left(\widehat{VaR}_\alpha(X) + \sum_{i=1}^{\lfloor m \rfloor} X_{(i)}\right)$$

ES estimator 3 and VaR estimator

$$\widehat{ES}_{\alpha}^{(3)}(X)$$

$$= -\frac{1}{\lfloor m \rfloor + 1}\left(\widehat{VaR}_{\alpha}(X) + \sum_{i=1}^{\lfloor m \rfloor - 1} X_{(i)}\right)$$

with

$$\widehat{VaR}_{\alpha}(X)$$

$$= -\left((1-(m-\lfloor m \rfloor))\, X_{(\lfloor m \rfloor)} + (m-\lfloor m \rfloor) X_{(\lfloor m \rfloor + 1)}\right).$$

Note that per construction $\widehat{ES}_{\alpha}^{(1)}(X) > \widehat{ES}_{\alpha}^{(2)}(X)$.

Contents

Normally distributed value changes

We start by assuming that the random change in value ΔV of our position V over the time horizon $[0, \Delta t]$ follows a normal distribution with parameters μ, ,

$$\Delta V \sim \mathrm{N}(\mu, \sigma^2). \tag{13}$$

A specification of driving risk factors is not required at this stage. According to Equation (11), $\mathbf{VaR}_{\alpha,\Delta t}$ is defined as

$$\begin{aligned} 1-\alpha &= \mathbf{P}(-\Delta V \geq \mathbf{VaR}_{\alpha,\Delta t}) \\ &= \mathbf{P}\left(\frac{-\Delta V + \mu}{\sigma} \geq \frac{\mathbf{VaR}_{\alpha,\Delta t} + \mu}{\sigma}\right) \\ &= 1 - \mathbf{P}\left(\frac{-\Delta V + \mu}{\sigma} < \frac{\mathbf{VaR}_{\alpha,\Delta t} + \mu}{\sigma}\right) \\ &= 1 - \mathrm{N}\left(\frac{\mathbf{VaR}_{\alpha,\Delta t} + \mu}{\sigma}\right), \end{aligned}$$

where N stands for the distribution function of the standard normal distribution $\mathrm{N}(0, 1)$.

Normally distributed value changes

Then,

$$\begin{aligned} N^{(-1)}\left(N\left(\frac{\mathbf{VaR}_{\alpha,\Delta t}+\mu}{\sigma}\right)\right) &= N^{(-1)}(\alpha) \\ \Rightarrow N_\alpha &= \frac{\mathbf{VaR}_{\alpha,\Delta t}+\mu}{\sigma} \\ \Rightarrow \mathbf{VaR}_{\alpha,\Delta t} &= \sigma N_\alpha - \mu \end{aligned}$$

Neglecting the drift

Denoting by N_p the p-quantile of the standard normal distribution yields

$$\mathbf{VaR}_{\alpha,\Delta t} = -\mathrm{N}_{1-\alpha}\sigma - \mu. \tag{14}$$

Since **VaR** is usually calculated for rather short time frames Δt such as 1 day or 10 days it is often justified to assume that the expected value change can be ignored, i.e., $\mu_{\Delta V} = 0$. We then have

$$\mathbf{VaR}_{\alpha,\Delta t} = -\mathrm{N}_{1-\alpha}\sigma, \tag{15}$$

and **VaR** is solely determined by the standard deviation σ of the value change ΔV and the normal quantile[1] N_α,

α	99,90%	99,50%	99,00%	95,00%	90,00%
N_α	3,0902	2,5758	2,3263	1,6449	1,2816

Remarks

1. We have normally distributed value changes ΔV if and only if the discrete return R^d of our position over the time horizon Δt possesses a normal distribution:

$$\Delta V = V_0 R^d.$$

If the return R^d follows an $\mathrm{N}(\mu_{R^d}, \sigma^2_{R^d})$ normal distribution, then

$$\Delta V \sim \mathrm{N}(V_0 \cdot \mu_{R^d}, V_0^2 \cdot \sigma^2_{R^d}).$$

In practice, one prefers to estimate parameters from historical return data and not from historical realized absolute value changes, the reason being that return estimates yield more stable results.

Remarks

2. Often daily historical data are used in practical applications, i.e., **VaR** is initially calculated referring to a 1 day time frame. For longer time horizons, e.g., $\Delta t = 10$ days, the one day **VaR** is scaled up according to the rule

$$\mathbf{VaR}_{\mathbf{10days}} = \mathbf{VaR}_{\mathbf{1day}}\sqrt{10}.$$

This is supported by the assumptions that the variance $\sigma^2_{\Delta V}$ over the interval $[0, \Delta t]$ in Equation (15) increases linearly with the length Δt of the time interval. If the expectation $\mu_{\Delta V}$ is included in the **VaR** calculation as in Equation (14) the expectation is linearly scaled.

The scaling rule is exact for iid log-returns:

$$R^c_{0,2} = \ln\left(\frac{V_2}{V_0}\right) = \ln\left(\frac{V_2}{V_1}\cdot\frac{V_1}{V_0}\right) = \ln\left(\frac{V_2}{V_1}\right) + \ln\left(\frac{V_1}{V_0}\right) = R^c_{1,2} + R^c_{0,1},$$

and hence $E(R^c_{0,2}) = 2\mu$, $\mathrm{Var}(R^c_{0,2}) = 2\,\mathrm{Var}(R^c) = 2\,\sigma^2$.

VaR of two positions

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Assume given two positions (portfolios) V^1, V^2 with jointly normally distributed changes $(\Delta V^1, \Delta V^2)$. How do the **VaR**'s of the individual positions V^1 and V^2 relate to the **VaR** for the aggregated position $V^1 + V^2$?
Assuming a correlation of ρ, the variance resp. standard deviation of the joint value change $\Delta V^1 + \Delta V^2$ is given by

$$\begin{aligned}
\sigma^2_{\Delta V^1 + \Delta V^2} &= \sigma^2_{\Delta V^1} + \sigma^2_{\Delta V^2} + 2\rho\sigma_{\Delta V^1}\sigma_{\Delta V^2} \\
\sigma_{\Delta V^1 + \Delta V^2} &\leq \sigma_{\Delta V^1} + \sigma_{\Delta V^2} \\
\sigma_{\Delta V^1 + \Delta V^2} &= \sigma_{\Delta V^1} + \sigma_{\Delta V^2} \qquad \text{if and only if } \rho = 1.
\end{aligned}$$

In view of $\mathrm{N}_{1-\alpha} < 0$ for $\alpha > 50\%$ this implies

$$\mathbf{VaR}^{1+2}_{\alpha,\Delta t} \leq \mathbf{VaR}^{1}_{\alpha,\Delta t} + \mathbf{VaR}^{2}_{\alpha,\Delta t} \tag{16}$$

$$\mathbf{VaR}^{1+2}_{\alpha,\Delta t} = \mathbf{VaR}^{1}_{\alpha,\Delta t} + \mathbf{VaR}^{2}_{\alpha,\Delta t} \qquad \text{if and only if } \rho = 1.$$

VaR of joint positions

As expected from a reasonable risk measure, the **VaR** of the joint position is bounded from above by the sum of the **VaR**s of the individual positions. In general, adding up **VaR** quantities over sub-positions overestimates the overall **VaR** of the aggregated position, unless they are perfectly correlated.
Regulators propose to aggregate risk figures by summation, which is clearly a conservative risk estimate.
However, in case of non-normally distributed value changes, as, for example in case of default risks, the inequality (16) can be violated. This is a serious argument against using **VaR** as a risk measure.

ES of normally distributed returns

Formula for expected shortfall assuming normally distributed value changes:

$$\mathbf{ES}_\alpha = -\mu + \sigma \, \frac{\mathrm{n}(\mathrm{N}_\alpha)}{1-\alpha}.$$

Derivation of ES formula: Assume that $\Delta V \sim \mathrm{N}(\mu, \sigma^2)$ and that ΔV denotes gains. Then, the conditional cumulative probability that a loss exceeds some value x, given that the loss (as a positive number) exceeds **VaR**, is

$$\begin{aligned}
&\mathbf{P}(-\Delta V \leq x | -\Delta V \geq \mathbf{VaR}_\alpha) \\
&= \frac{\mathbf{P}(\mathbf{VaR}_\alpha \leq -\Delta V \leq x)}{1-\alpha} \\
&= \mathbf{P}\left(\frac{\mathbf{VaR}_\alpha + \mu}{\sigma} \leq \frac{-\Delta V + \mu}{\sigma} \leq \frac{x+\mu}{\sigma}\right) / (1-\alpha) \\
&= \left[\mathrm{N}\left(\frac{x+\mu}{\sigma}\right) - \mathrm{N}\left(\frac{\mathbf{VaR}_\alpha + \mu}{\sigma}\right)\right] \mathbf{1}_{\{x \geq \mathbf{VaR}_\alpha\}} / (1-\alpha)
\end{aligned}$$

The above conditional cumulative probability can be used to verify that the conditional density is $f_{con}(x) = \mathrm{n}\left(\frac{x+\mu}{\sigma}\right) \cdot \frac{1}{\sigma(1-\alpha)} \mathbf{1}_{\{x \geq \mathbf{VaR}_\alpha\}}$

Proof:

$$
\begin{aligned}
&\mathbf{P}(-\Delta V \leq x | -\Delta V \geq \mathbf{VaR}_\alpha) \\
&= \mathbf{1}_{\{x \geq \mathbf{VaR}_\alpha\}} \int_{-\infty}^{x} f_{con}(y)dy \\
&= \mathbf{1}_{\{x \geq \mathbf{VaR}_\alpha\}} \int_{-\infty}^{x} \mathrm{n}\left(\frac{y+\mu}{\sigma}\right) \cdot \frac{1}{\sigma(1-\alpha)} \mathbf{1}_{\{y \geq \mathbf{VaR}_\alpha\}} dy \\
&= \mathbf{1}_{\{x \geq \mathbf{VaR}_\alpha\}} \int_{-\infty}^{\frac{x+\mu}{\sigma}} \mathrm{n}(z) \frac{1}{\sigma(1-\alpha)} \mathbf{1}\underbrace{\{\sigma z - \mu \geq \mathbf{VaR}_\alpha\}}_{\iff z \geq \frac{\mathbf{VaR}_\alpha + \mu}{\sigma}} \underbrace{\mathrm{d}(\sigma z - \mu)}_{= \sigma dz} \\
&= \frac{\mathbf{1}_{\{x \geq \mathbf{VaR}_\alpha\}}}{1-\alpha} \int_{\frac{\mathbf{VaR}_\alpha + \mu}{\sigma}}^{\frac{x+\mu}{\sigma}} \mathrm{n}(z)\,\mathrm{d}z \\
&= \frac{\mathbf{1}_{\{x \geq \mathbf{VaR}\}}}{1-\alpha} \left[\mathrm{N}\left(\frac{x+\mu}{\sigma}\right) - \mathrm{N}\left(\frac{\mathbf{VaR}_\alpha + \mu}{\sigma}\right)\right]
\end{aligned}
$$

ES of normally distributed returns

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Expected shortfall:

$$
\begin{aligned}
&\mathbb{E}\left[-\Delta V \mid -\Delta V \geq \mathbf{VaR}_\alpha\right] \\
&= \int_{-\infty}^{\infty} x \mathrm{n}\left(\frac{x+\mu}{\sigma}\right) \frac{1}{\sigma(1-\alpha)} \mathbf{1}_{\{x \geq \mathbf{VaR}_\alpha\}} \,\mathrm{d}x \\
&= \int_{-\infty}^{\infty} (\sigma y - \mu) \frac{\mathrm{n}(y)}{\sigma(1-\alpha)} \mathbf{1}_{\left\{y \geq \frac{\mathbf{VaR}_\alpha + \mu}{\sigma}\right\}} \mathrm{d}(\sigma y - \mu) \\
&= \int_{\frac{\mathbf{VaR}_\alpha+\mu}{\sigma}}^{\infty} (\sigma y - \mu) \frac{\mathrm{n}(y)}{1-\alpha} \,\mathrm{d}y \\
&= \int_{\frac{\mathbf{VaR}_\alpha+\mu}{\sigma}}^{\infty} \sigma y \frac{\mathrm{n}(y)}{1-\alpha} \,\mathrm{d}y - \int_{\frac{\mathbf{VaR}_\alpha+\mu}{\sigma}}^{\infty} \mu \frac{\mathrm{n}(y)}{1-\alpha} \,\mathrm{d}y \\
&= \frac{\sigma}{1-\alpha} \int_{\frac{\mathbf{VaR}_\alpha+\mu}{\sigma}}^{\infty} y \mathrm{n}(y) \,\mathrm{d}y - \frac{\mu}{1-\alpha} \int_{\frac{\mathbf{VaR}_\alpha+\mu}{\sigma}}^{\infty} \mathrm{n}(y) \,\mathrm{d}y \\
&= \frac{\sigma}{1-\alpha} \int_{\frac{\mathbf{VaR}_\alpha+\mu}{\sigma}}^{\infty} y \mathrm{n}(y) \,\mathrm{d}y - \frac{\mu}{1-\alpha} \left(1 - \underbrace{N\left(\frac{\mathbf{VaR}_\alpha + \mu}{\sigma}\right)}_{=\alpha} \right)
\end{aligned}
$$

ES of normally distributed returns

Observe that it holds for the standard normal density $n(x)$

$$n'(y) = -y\mathrm{n}(y)$$

so that, continuing above

$$\begin{aligned}
&\mathbb{E}\left[-\Delta V \mid -\Delta V \geq \mathbf{VaR}_\alpha\right] \\
&= \frac{-\sigma}{1-\alpha} \int_{\frac{\mathbf{VaR}_\alpha + \mu}{\sigma}}^{\infty} n'(y)\,\mathrm{d}y - \mu \\
&= \frac{-\sigma}{1-\alpha} n(y) \Big|_{\frac{\mathbf{VaR}_\alpha + \mu}{\sigma}}^{\infty} - \mu \\
&= \frac{\sigma}{1-\alpha} n\left(\frac{\mathbf{VaR}_\alpha + \mu}{\sigma}\right) - \mu \\
&= \frac{\sigma}{1-\alpha} \mathrm{n}\left(\mathrm{N}_\alpha\right) - \mu.
\end{aligned}$$

- We calculate VaR for a simple example taken from credit risk.

Example 1

- We consider 500 independent loans each with notional amount of 100.
- The probability of default of each borrower over the time horizon Δt is 2%.
- The default of loan i is described by the r.v. $Z_i = \begin{cases} 1, & \text{if i defaults,} \\ 0, & \text{otherwise,} \end{cases}$

  with $\mathbf{P}(Z_i = 1) = 2\%$.
- The loss $L_i = -\Delta V_i$ of the position in loan i is $L_i = 100 \cdot Z_i$.
- The overall loss is $L = \sum_{i=1}^{500} L_i = 100 \sum_{i=1}^{500} Z_i = 100\,\xi$, with $\xi = \sum_{i=1}^{500} Z_i$.
- The r.v. ξ admits a Binomial distribution $B(n, p)$ with $n = 500$, $p = 2\%$.
- The 95% quantile of ξ is 15,[2] yielding a **VaR** of $\mathbf{VaR}_{\alpha,\Delta t}(L) = 1500$. [..]

[2] $\mathbf{P}(\xi \leq 14) = 0.9187$ and $\mathbf{P}(\xi \leq 15) = 0.953$

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Example 2 (cont'd)

- On the other hand, the loss L_i for each sub position is distributed according to $\mathbf{P}(L_i = 0) = 98\%$, $\mathbf{P}(L_i = 100) = 2\%$.
- This implies for each single loan a **VaR** of $\mathbf{VaR}_{\alpha,\Delta t}(L_i) = 0$.
- This implies that the risk of the portfolio is greater than the risk of the sum of the individual positions:

$$1500 = \mathbf{VaR}_{\alpha,\Delta t}(L) > \sum_{i=1}^{500} \mathbf{VaR}_{\alpha,\Delta t}(L_i) = 0.$$

- However, a reasonable risk measure should produce the opposite, namely that that the overall risk is smaller if the portfolio is diversified!
- Moreover, a single loan with a face value of $500 \cdot 100$ receives a **VaR** of zero, whereas an obviously diversified portfolio of 500 independent loans each with notional of 100 obtains a **VaR** of 1500.

[..]

Example

- That **VaR** of a portfolio may be greater than the sum of the individual **VaR**'s has to do with the failure of **VaR** to take into account the severity of losses below **VaR**.
- The risk measure **expected shortfall** overcomes this drawback.

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Example 3 (cont'd)

- We calculate **ES**:

$$\mathbf{ES}_{\alpha,\Delta t}(L) = \frac{500 \cdot 100\, P(\xi = 500) + \cdots + 1600\, \mathbf{P}(\xi = 16)}{1-\alpha} + \frac{1500 \cdot (\mathbf{P}(\xi \leq 15) - \alpha)}{1-\alpha}$$
$$= 1606.5 + \frac{1500 \cdot (0.953 - 0.95)}{0.05} = 1696.5,$$

  and for each single loan i

$$\mathbf{ES}_{\alpha}(L_i) = \frac{100\, \mathbf{P}(Z_i = 1) + 0(\mathbf{P}(L_i \leq 0) - \alpha)}{1-\alpha} = 40.$$

- The portfolio **ES** is significantly smaller than the sum of the individual **ES**'s:

$$1696.5 = \mathbf{ES}_{\alpha}(\Delta V) \leq \sum_{i=1}^{500} \mathbf{ES}_{\alpha}(\Delta V_i) = 20\,000.$$

Contents

Axioms for risk measures

- This deficiency of **VaR** has led to the formulation of a set of axioms that any risk measure should satisfy.
- A **risk measure** is a **function** that assigns the **loss** $L = -\Delta V$ associated with a position V a risk number $\varrho(L)$.[3]
- We **interpret** $\varrho(L)$ as the amount of capital that should be added to a position with associated loss L, so that the position becomes **acceptable** to an external or internal risk controller.
- Positions with $\varrho(L) \leq 0$ are acceptable without injection of capital; if $\varrho(L) < 0$, capital may even be withdrawn.
- We now specify the four axioms.

[3] We follow the notation of McNeil et al. ?.

Axiom 5.1 (Translation invariance, cash invariance)

For any fixed number $l \in \mathbb{R}$ we have

$$\varrho(L - l) = \varrho(L) - l.$$

- Interpretation: If a given position with loss L is supplemented by a capital amount of $l > 0$, then the risk of the combined position $\varrho(L - l)$ is just the risk $\varrho(L)$ lowered by the amount l.
- The axiom is necessary for the risk-capital interpretation above: adding the capital $\varrho(L)$ gives $\varrho(L - \varrho(L)) = \varrho(L) - \varrho(L) = 0$; in words, adding $\varrho(L)$ to the position L makes the position acceptable.

Axioms for risk measures

Axiom 5.2 (Subadditivity)

For positions with associated losses L_1, L_2 we have

$$\varrho(L_1 + L_2) \leq \varrho(L_1) + \varrho(L_2).$$

- Interpretation: The risk $\varrho(L_1 + L_2)$ of a combined position does not exceed the sum of the individual risks.
- The motivation for this axiom consists of the following arguments:
  - Risk can be lowered by diversification.
  - A non-subadditive risk measure suggests that there are situations where it is favorable to split a position to decrease the risk and the associated regulatory capital.
  - A subadditive risk measure is the precondition for a non-centralized risk management: to ensure that the overall risk, $\varrho(L_1 + L_2)$, is bounded by a barrier C, it suffices to limit the individual risks $\varrho(L_i)$ by bounds C_i with $C_1 + C_2 \leq C$.

Axiom 5.3 (Positive homogeneity)

For any factor $\lambda \geq 0$ we have

$$\varrho(\lambda L) = \lambda \varrho(L).$$

- Interpretation: The risk of a positive multiple of a position is the multiple of the risk of the initial position.
- As a consequence of subadditivity, we already have that for any $n \in \mathbb{N}$

$$\varrho(nL) = \varrho(L + \cdots + L) \leq n\varrho(L).$$

Since there is no diversification between the losses of the portfolio, it is natural to require that equality should hold in this case.

Axiom 5.4 (Monotonicity)

For any positions with associated losses L_1, L_2 such that $L_1 \leq L_2$ we have $\varrho(L_1) \leq \varrho(L_2)$.

- Interpretation: Positions that lead to higher losses in every state of the world require more risk capital.

Exercise 1

Show that for a risk measure satisfying the axioms, the risk of a position is bounded by the maximum loss.

Exercise 1

Show that for a risk measure satisfying the axioms, the risk of a position is bounded by the maximum loss.

Solution:

- For a position L observe that

$$L \leq \sup L,$$

  where $\sup L$ denotes the supremum (maximum) loss.
- Axiom 4 implies:

$$\varrho(L) \leq \varrho(\sup L).$$

- Because $\sup L$ is a number, it follows by Axiom 1 (cash invariance):

$$\varrho(\sup L) \overset{\text{cash\_inv.}}{=} \sup L.$$

- Hence, $\varrho(L) \leq \sup L$.

Definition 1

A risk measure $\varrho(\Delta V)$ satisfying axioms 5.1–5.4 is called a **coherent risk measure**.

Remarks:

- The Axiom of positive homogeneity has received criticism; it has been suggested that for large values of λ, one should have $\varrho(\lambda L) > \lambda\varrho(L)$, reflecting the concentration risk and possibly liquidity problems in unwinding the position.
- This has lead to the definition of the larger class of **convex risk measures**.
- A treatment is beyond the scope of the course.

Contents

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VaR and subadditivity

- From the definition of **VaR**, Equation (12), one can show that the axioms of positive homogeneity, translation invariance and monotonicity are satisfied (Exercise!).
- For **normally distributed value changes** the **subadditivity** of **VaR** holds (see previous chapter).
- However, in the general situation **VaR** is **not coherent** since the axiom of subadditivity can be violated. This is a serious argument against using **VaR** in practice!

Contents

Simulation

Simulation approaches to **VaR** are based on a representative set of possible realizations of the random value change ΔV. Our starting point is again a factor mapping explaining the value changes by factor changes ΔY: $\Delta V = \Delta V(\Delta Y)$. For concrete factor changes $\Delta \mathbf{y}_1, \dots, \Delta \mathbf{y}_k$ we calculate the corresponding realized value changes $\Delta V(\Delta \mathbf{y}_i)$. Now **VaR** is estimated from these realized values $\Delta V(\Delta \mathbf{y}_1), \dots, \Delta V(\Delta \mathbf{y}_k)$ by computing their **empirical quantile** $\hat{Q}_{1-\alpha}$. The empirical quantile for level $(1-\alpha)$ is the $[(1-\alpha) \cdot k + 1]$-smallest value out of $\Delta V(\Delta \mathbf{y}_1), \dots, \Delta V(\Delta \mathbf{y}_k)$. By the law of large numbers we know that the empirical quantile approximates (converges to) the true quantile, so

$$\mathbf{VaR}_{\alpha,\Delta t} \approx -\hat{Q}_{1-\alpha}.$$

Drawbacks of simulation approaches are the problem of **getting representative simulations** and the **demanding numerical efforts**, since for each simulation scenario $\Delta \mathbf{y}_i$ the implied value change $\Delta V(\Delta \mathbf{y}_i)$ requires a complete revaluation of the underlying position.

The latter problem can be relaxed by re-evaluating the underlying position only approximately. Again, for a first order approximation one can use for example the linearization as in the delta normal approach above:

$$\Delta V(\Delta \mathbf{y}) \approx \frac{\partial}{\partial t} f(0, Y_0)\Delta t + \sum_{j=1}^{d} \frac{\partial}{\partial y_j} f(0, \mathbf{Y}_0)\Delta \mathbf{y}^j.$$

By applying Taylor's formula to some higher order one can improve the quality of the approximation. For example, in case of an option position $V_t = f(t, S_t, r_t, \sigma_t)$ with strong non-linearity in the price S_t of the underlying, a common technique in practice is to use the following **Delta-Gamma-Vega-Theta** approximation

$$\Delta V \approx \frac{\partial}{\partial t} f(0, S_0, r, \sigma_0)\Delta t + \frac{\partial}{\partial S_0} f(0, S_0, r, \sigma_0)\Delta S + 1/2\frac{\partial^2}{\partial S_0^2} f(0, S_0, r, \sigma_0)(\Delta S)^2 + \frac{\partial}{\partial \sigma_0} f(0, S_0, r, \sigma_0)\Delta\sigma.$$

Contents

Historical Simulation

For historical simulation, the simulation scenarios are simply taken from historically observed factor changes $\Delta\mathbf{y}_1, \dots, \Delta\mathbf{y}_k$. There is no assumption about the probability distribution of the factor changes needed. Dependencies are automatically taken into account by the historical data - of course one has to use **joint historical factor changes** for all factors involved!

The following **problems** appear when using historical simulations:

- The appropriate number k of past simulations is not easy to define. On the one hand a long history with many data points yields a tighter estimate of the quantile. On the other hand it is questionable whether data points from the far past are still representative for the distribution of future factor changes.
- Sometimes sufficient historical data are not available for all risk factors. Moreover, the data has to be collected synchronized in order to represent dependencies in an unbiased way.
- Very long historical time series may contain too many extreme events, so that application to the current position overestimates the true risk.
- Conversely, it is also possible that the data series contains too few extreme scenarios.

Contents

Monte Carlo simulation

For Monte Carlo simulation, the simulation scenarios $\Delta \mathbf{y}_1, \dots, \Delta \mathbf{y}_k$ for the factor changes are generated artificially by a computer. This requires **explicit assumptions on the joint distribution** of the random vector $(\Delta Y^1, \dots, \Delta Y^d)$ of **changes of the d risk factors**. The joint distribution involves the marginal distribution of each single factor change ΔY^j as well as their dependencies, correlations. In fact, the **main problem of Monte Carlo simulation** is the determination of an appropriate joint distribution.
Monte Carlo simulation can be realized easily if one assumes a d-dimensional normal distribution for the joint factor changes,

$$\Delta \mathbf{Y} = (\Delta Y^1, \dots, \Delta Y^d)^T \sim \mathrm{N}(\mu_{\Delta \mathbf{Y}}, \mathrm{Cov}(\Delta \mathbf{Y})),$$

with expectation vector $\mu_{\Delta \mathbf{Y}}$ and covariance matrix $\mathrm{Cov}(\Delta \mathbf{Y})$.

Monte Carlo simulation

In practice one has to estimate $\mu_{\Delta \mathbf{Y}}$ and $\mathrm{Cov}(\Delta \mathbf{Y})$ again from historical data.

- Advantage of simulation of Delta-normal approach: can apply simulation this to non-linear risk mappings (e.g. option positions)
- Advantage of MC simulation over historical simulation: more simulations possible.
- MC simulation (risk factors in parts modelled as t-distributed, possibly with Gaussian copula) industry standard
- Some banks use historical simulation

In addition to the above mentioned problem of choosing the right distribution for the changes of the risk factors we face the following **problems when using a Monte-Carlo simulation** approach:

- The number k of required simulations is usually quite high in order to guarantee a stable **VaR** result. Stable in this context means that repeated **VaR** calculations with newly generated sets of simulations return similar **VaR** results. So-called variance reduction techniques help to stabilize the final results of the simulation.
- The number d of risk factors involved is often quite large and one is forced to reduce the number of factors to take only the most important risk factors into account. The technique of principal components analysis detects those risk factors that contribute most to the variability of the value of the position.

Contents

Copulas and dependence

- We turn to modelling the dependence among several risk factors.
- The analysis focusses on the bivariate case; extensions to multivariate models are straight-forward, but notationally more demanding.
- Every joint distribution function for two risk factors describes both the **marginal behaviour** of the individual risk factors and the **dependence structure**.
- The principal idea of the **copula** approach is to isolate the description of the dependence structure.
- Consequently, copulas allow for a **bottom-up approach** to multivariate model building, by separately modelling the marginal distributions and the dependence.

Contents

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- Let $(x_1, y_1), \dots, (x_n, y_n)$ be a sample of joint observations.
- The **sample covariance** is defined as

$$\hat{\sigma}_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n-1}\sum_{i=1}^{n} x_i y_i - \frac{n}{n-1}\bar{x}\bar{y}.$$

- The **sample correlation coefficient** is defined as

$$\hat{\rho}_{xy} = \frac{\sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}}{\sqrt{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2} \cdot \sqrt{\sum_{i=1}^{n} y_i^2 - n\bar{y}^2}} = \frac{\hat{\sigma}_{xy}}{\hat{\sigma}_x \hat{\sigma}_y},$$

- Interpretation: $\hat{\rho}_{xy}$ measures the degree of **linear dependence** of two variables X and Y

Correlation coefficient: Interpretation

Properties of the sample correlation coefficient:

- $-1 \leq \hat{\rho}_{xy} \leq 1$
- $\rho > 0$: Positive linear relationship
- $\rho < 0$: Negative linear relationship
- $\rho = 0$: uncorrelated; no linear relationship.

Linear correlation

- To motivate the study of copulas further, we first illustrate why **correlation** is not a suitable measure to fully capture the dependence among random variables.
- Correlation as a measure of dependence has a number of limitations:
  - If X and Y are independent, then $\rho(X, Y) = 0$, but the converse it **not true** in general ($\rightsquigarrow$ Ex. 2).
  - The marginal distributions and pairwise correlations of a random vector **do not** determine its joint distribution ($\rightsquigarrow$ Ex. 3)

Linear correlation

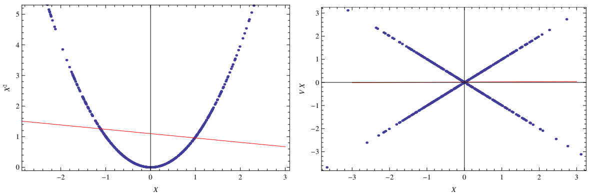

Figure: Scatterplots of the variables introduced in Exercises 2 (left), 3 (right). Each scatterplot consists of 500 randomly generated samples. Correlation coefficients are: 0.01, 0, respectively.

Exercise 2

Let X be standard normally distributed. Determine the correlation of X and X^2. Are X and X^2 independent? (See also Figure above.)

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Exercise 3

Let X be a standard normally distributed random variable and let V be an independent random variable with $\mathbf{P}(V = 1) = \mathbf{P}(V = -1) = 0.5$. Set $Y = VX$. Show that Y is also normally distributed and that the correlation between X and Y is 0. Further show that X and Y are not independent.

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Solution of **Exercise 2**:

- Correlation of X and X^2: $\rho(X, X^2) = \mathbb{E}(X \cdot X^2) = \mathbb{E}(X^3) = 0$, since the normal distribution is symmetric and thus has skew 0.
- But $\mathbf{P}(X \leq 1, X^2 \leq 1) = \mathbf{P}(X^2 \leq 1) \neq \mathbf{P}(X \leq 1)\mathbf{P}(X^2 \leq 1)$, so they are not independent.

Solution of **Exercise 3**:

- $Y = VX$ normally distributed:

$$\begin{aligned}\mathbf{P}(VX \leq x) &= \mathbf{P}(X \leq x, V = 1) + \mathbf{P}(-X \leq x, V = -1) \\ &= \mathbf{P}(X \leq x) \cdot \mathbf{P}(V = 1) + \mathbf{P}(-X \leq x) \cdot \mathbf{P}(V = -1) \\ &= \mathrm{N}(x) \cdot \frac{1}{2} + \mathrm{N}(x) \cdot \frac{1}{2}.\end{aligned}$$

- Correlation: $\mathbb{E}(X \cdot VX) = \mathbb{E}(V) \cdot \mathbb{E}(X^2) = 0 \cdot 1 = 0$.
- Joint distribution at $x = y = 1$:

$$\begin{aligned}\mathbf{P}(X \leq 1, VX \leq 1) &= \mathbf{P}(X \leq 1, V = 1) + \mathbf{P}(X \leq 1, -X \leq 1, V = -1) \\ &= \frac{1}{2}\mathrm{N}(1) + \frac{1}{2}(\mathrm{N}(1) - \mathrm{N}(-1)) = \mathrm{N}(1) - \frac{1}{2}\mathrm{N}(-1) \\ &\neq \mathrm{N}(1)^2 = \mathbf{P}(X \leq 1) \cdot \mathbf{P}(VX \leq 1).\end{aligned}$$

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Definition 2

A (bivariate) **copula** is a distribution function on $[0,1]^2$ with standard uniform marginals.

- Loosely speaking, the set of bivariate copulas comprises bivariate distribution functions that differ only in the dependence structure, not in the marginals.
- We write $C(u,v) = \mathbf{P}(U \leq u, V \leq v)$, where $U, V \sim U(0,1)$ for copulas.
- Any copula satisfies the following two properties
  (i) $C(0,u) = C(u,0) = 0$ and $C(1,u) = C(u,1) = u$, for all $u \in [0,1]$,
  (ii) For any $a \leq b$ and $c \leq d$, we have
  $\mathbf{P}(U \in [a,b], V \in [c,d]) = C(b,d) - C(a,d) - C(b,c) + C(a,c) \geq 0.$
- Conversely, any function $C : [0,1]^2 \to [0,1]$ satisfying these two properties is a copula.

Definition

- The first property on the previous page can be well understood by remembering that $C(0, v) = \mathbf{P}(U \leq 0, V \leq v) = 0$ as $\mathbf{P}(U \leq 0) = 0$ given that $U \sim U[0, 1]$.
- The second one can be motivated as follows: For any $a \leq b$ and $c \leq d$ it holds:

$$
\begin{aligned}
0 &\leq \mathbf{P}(U \in [a, b], V \in [c, d]) \\
&= \mathbf{P}(U \leq b, U \geq a, V \leq d, V \geq c) \\
&= \mathbf{P}(U \leq b, V \leq d) - \mathbf{P}(U \leq a, V \leq d) \\
&\quad - \mathbf{P}(U \leq b, V \leq c) + \mathbf{P}(U \leq a, V \leq c) \\
&= [C(b, d) - C(a, d)] - [C(b, c) - C(a, c)]
\end{aligned}
$$

  This implies that $[C(b, d) - C(a, d)] \geq [C(b, c) - C(a, c)]$, which defines the shape of a copula function: The slope of its surface will increase the further we proceed on the y-axis!

Definition

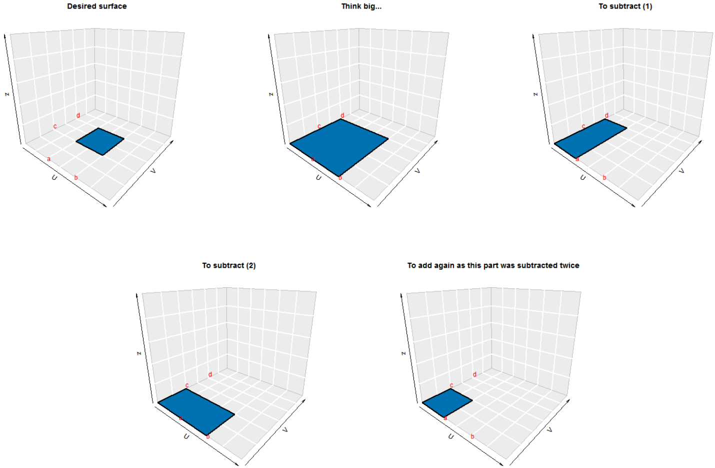


Figure: Graphical illustration of copula property (ii) of page 101.

Transforms on marginals

- The usefulness of copulas lies in the possibility to transform uniform marginals to other distributions and vice versa; in this way, the dependence structure and the marginals can be modelled separately:[4]
- If $U \sim U(0,1)$ and G is a distribution function, then $G^{(-1)}(U)$ follows the distribution G. Formally:

$$\mathbf{P}(G^{(-1)}(U) \leq x) = G(x), \qquad \text{for } x \in \mathbb{R}.$$

  In words: a uniformaly distributed random variable U can be transformed into a G-distributed random variable via $G^{(-1)}(U)$.
- If Y has distribution function G, then $G(Y) \sim U(0,1)$. Formally:

$$P(G(Y) \leq u) = \begin{cases} 0, & \text{for } u < 0, \\ u, & \text{for } u \in [0,1], \\ 1, & \text{for } u > 1. \end{cases}$$

  In words: a G-distributed random variable Y is transformed into a uniform random variable by $G(Y)$.

[4] All distributions are assumed to be continuous; the extension to discrete distributions is straightforward, but requires separate notation.

Illustration of marginal transformation

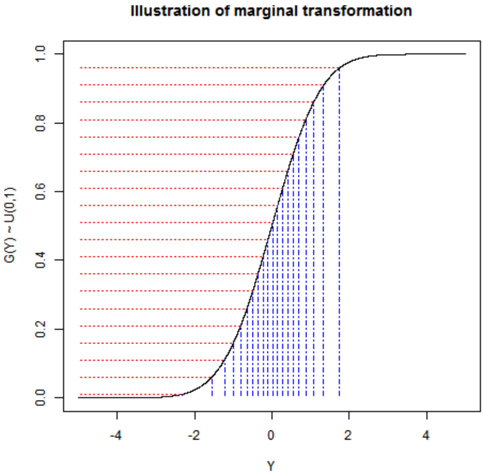


Figure: Marginal transformation.

Sklar's Theorem

- The importance of copulas is captured by Sklar's Theorem:

Theorem 3 (Sklar's Theorem)

Let F be a joint distribution function with margins F_1, F_2. Then, there exists a copula $C : [0,1]^2 \to [0,1]$ such that, for all $x, y \in \mathbb{R}$

$$F(x,y) = C(F_1(x), F_2(y)). \tag{17}$$

If the margins are continuous, then C is unique; otherwise C is unique on the range of the margins.
Conversely, if C is a copula and F_1, F_2 are univariate distribution functions, then the function F defined by Equation (17) is a joint distribution function with margins F_1, F_2.

- An explicit representation of C in terms of F and its margins is given by
$$C(u,v) = F(F_1^{(-1)}(u), F_2^{(-1)}(v)).$$
- Hence with any pair of random variables (X, Y) with joint distribution function F and marginals F_1, F_2 is associated a copula C of $(F_1(X), F_2(Y))$.

Fréchet-Hoeffding bounds and product copula

- Every copula C lies within the **Fréchet-Hoeffding bounds**:

$$\max(u + v - 1, 0) \leq C(u, v) \leq \min(u, v).$$

- The lower bound is the **countermonotonicity copula**, which is the copula of random variables X and Y that are decreasing functions of one another.
- The upper bound is the **comonotonicity copula**, which is the copula of random variables X and Y that are increasing functions of one another.
- The **product copula** $C(u, v) = uv$, which is the copula of **independent** random variables.

Fréchet-Hoeffding bounds and product copula

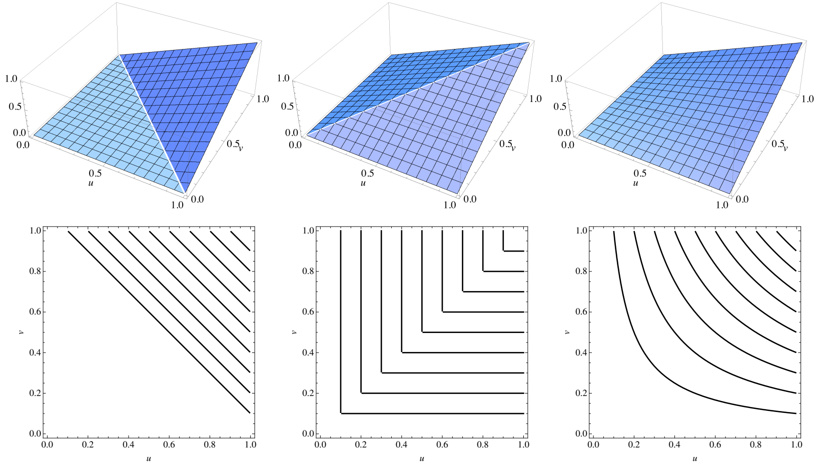


Figure: Perspective and contour plots of the countermonotonicity copula (left), comonotonicity copula (middle) and product (independence) copula (right).

- Note that the copula function C discussed on the previous slides is a cumulative distribution function, assuming values in $[0, 1]$.
- Taking this teaching approach was important to first understand the linkage between the marginal (cumulative) distributions $F1, F2$, the joint (cumulative) distribution F and the (cumulative) copula function.
- Simulating copula random numbers, however, requires copula density functions, i.e., the derivative of C.
- Examples are illustrated graphically on the next pages.

Copula examples

- A popular example is the **Gaussian copula**, also **Normal copula**, which is the copula generated by jointly normally distributed random variables:

$$C_{\rho,\mathrm{N}}(u,v) := \mathrm{N}_2(\mathrm{N}^{(-1)}(u), \mathrm{N}^{(-1)}(v); \rho).$$

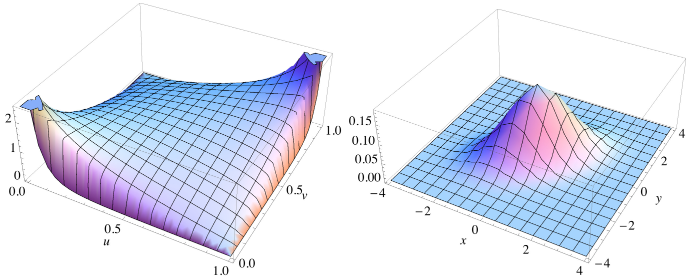


Figure: Left: Density of the Gaussian (Normal) copula. Right: Density of the bivariate Normal distribution ($\rho = 0.5$ in both cases).

Linkage between the two graphs on the previous page

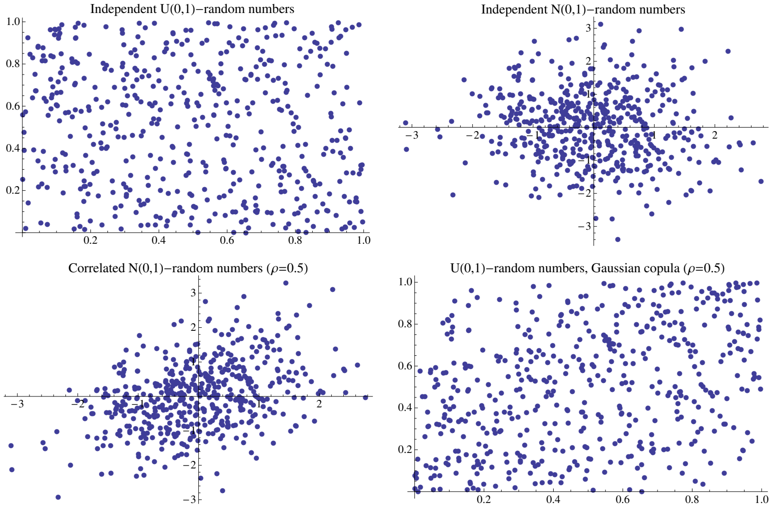
Dependency structure between x and y:

- The copula density assumes very high values around $[0,0]$ and $[1,1]$, whereas the opposite is true around $[0,1]$ and $[1,0]$.
- In other words: It is more likely that x and y are either very small at the same time or very large at the same time - and less likely that x is large, when y is small (respectively: x is small, when y is large). This is what we expect from correlated random variables!

Implied probability distribution:

- Note that the copula density is not too small around $[0.5, 0.5]$ and that at the same time the normal cumulative distribution function is very steep around 0.5.
- This explains the resulting high probability that the two-dimensional realisations of the bivariate normal distribution are close to [0,0].

Gaussian copula



Gaussian copula

- In a next step one can now transform the margins to an arbitrary distribution.
- This is the key to building models where the dependence is modelled separately from the marginals.
- A popular example in credit risk is to model joint default times (each exponentially distributed) with a Gaussian copula.

t-copula

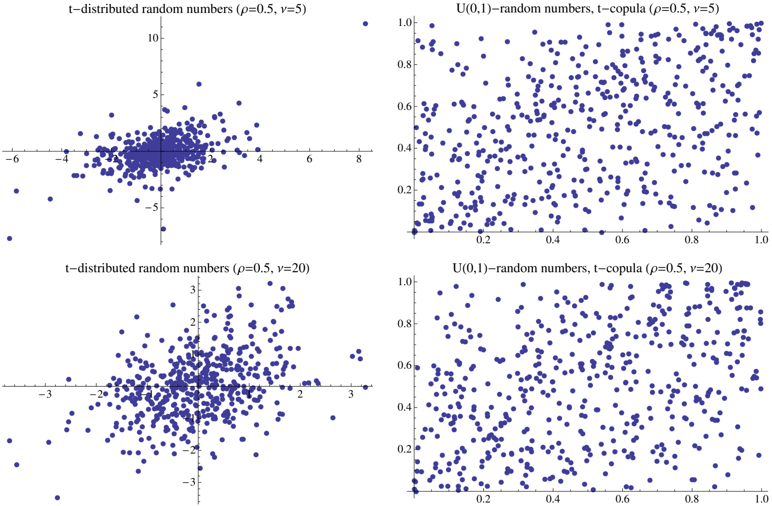
- The t-copula is the copula derived from a multivariate t-distribution:

$$C_{\nu,\rho,t} = t_{\nu,2}(t_\nu^{(-1)}(u), t_\nu^{(-1)}(v); \rho),$$

where

  - $t_{\nu,2}(x, y; \rho)$ is the bivariate t distribution function with degrees of freedom parameter ν and correlation parameter ρ;
  - t_ν is the univariate t distribution function.

t-copula: Examples



Archimedean copulas

- The copulas studied so far were generated from well-known multivariate distribution functions.
- A different approach is to generate copulas directly.
- A well-studied one-parameter family of copulas are the **Archimedean copulas**.
- Let $\phi : [0, 1] \to [0, \infty]$ be a continuous and strictly decreasing function with $\phi(1) = 0$ and $\phi(0) \leq \infty$.
- We define the **pseudo-inverse** of ϕ as

$$\phi^{(-1)}(t) = \begin{cases} \phi^{-1}(t), & 0 \leq t \leq \phi(0), \\ 0, & \phi(0) < t \leq \infty. \end{cases}$$

- If, in addition, ϕ is convex, then the following function is a copula:

$$C(u, v) = \phi^{(-1)}(\phi(u) + \phi(v)).$$

- Such copulas are called **Archimedean copulas**, and the function ϕ is called an **Archimedean copula generator**.

Gumbel and Clayton copulas

- Examples of Archimedean copulas are the **Gumbel** and the **Clayton** copulas:

$$C_{\theta,\mathrm{Gu}}(u,v) = \exp\left\{-((-\ln u)^{\theta} + (-\ln v)^{\theta})^{1/\theta}\right\}, \qquad 1 \le \theta < \infty,$$
$$C_{\theta,\mathrm{Cl}}(u,v) = (u^{-\theta} + v^{-\theta} - 1)^{-1/\theta}, \qquad 0 < \theta < \infty.$$

- In the case of the Gumbel copula, the independence copula is attained when $\theta = 1$ and the comonotonicity copula is attained as $\theta \to \infty$.
- Thus, the Gumbel copula interpolates between independence and perfect dependence.
- In the case of the Clayton copula, the independence copula is attained as $\theta \to 0$, whereas the comonotonicity copula is attained as $\theta \to \infty$.

Exercise 4

Determine ϕ and ϕ^{-1} of the Gumbel and Clayton copulas.

Gumbel and Clayton copulas

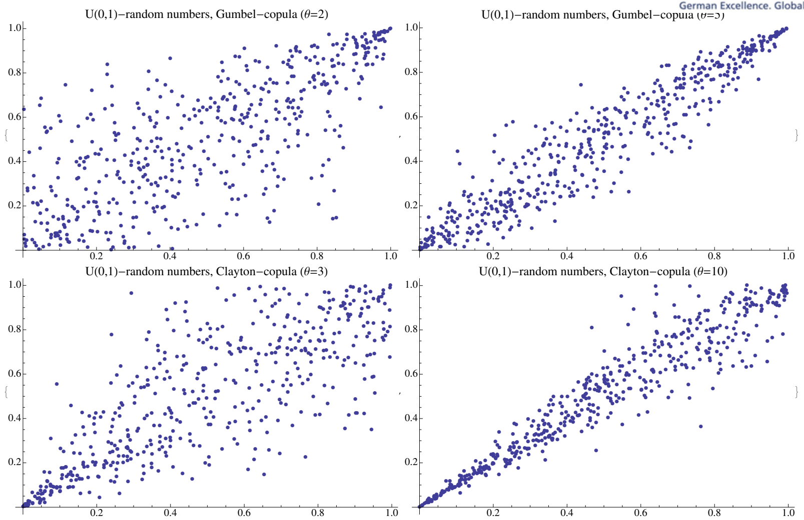


Figure: Random numbers from the Gumbel (top) and Clayton (bottom) copulas.

Dependence measures

- Linear correlation, as discussed earlier, depends on the specification of both the marginals and the joint distribution function (cf. Exercises 2 and 3).
- We introduce two measures of dependence that depend only on the copula and not on the marginal distributions: Spearman's rho and tail dependence.

Spearman's rho

- For random variables X, Y with marginals F_1, F_2, **Spearman's rho** is defined as
$$\rho_S(X,Y) = \rho(F_1(X), F_2(Y)),$$
- Aside from the independence of the marginals, one can show that Spearman's rho takes values in $[-1,1]$ with -1 corresponding to the countermonotonicity copula and 1 corresponding to the comonotonicity copula.
- In case of the independence copula, Spearman's rho takes the value 0 (although the converse statement does not necessarily hold).

Tail dependence

- An important dependence measure in risk management is the **tail dependence**, which provides a measure of extremal dependence by measuring the degree of dependence in the tails of a bivariate distribution.
- In the case of **upper tail dependence** we look at the probability that Y exceeds its q-quantile given that X exceeds its q-quantile and then consider the limit as q goes to 1. Formally:

$$\lambda_u := \lim_{q \uparrow 1} \mathbf{P}(Y > F_2^{(-1)}(q) | X > F_1^{(-1)}(q)).$$

- Correspondingly, we define the **lower tail dependence** to be

$$\lambda_l := \lim_{q \downarrow 0} \mathbf{P}(Y \leq F_2^{(-1)}(q) | X \leq F_1^{(-1)}(q)).$$

- These are properties of the copula: for example, the coefficient of lower tail dependence can be expressed in terms of the copulas as $\lambda_l = \lim_{q \downarrow 0} \dfrac{C(q,q)}{q}$.

Tail dependence

- One can show that the tail dependence (upper and lower) of the Gaussian copula is 0; in other words, random variables with a Gaussian copula are **asymptotically independent**.
- This makes it highly questionnable whether the Gaussian copula is appropriate for modelling dependence in risk management!
- The t-copula is asymptotically dependent in both the upper and lower tail provided that $\rho > -1$.
- The Gumbel copula has upper tail dependence while the Clayton copula has lower tail dependence.

Statistical tests

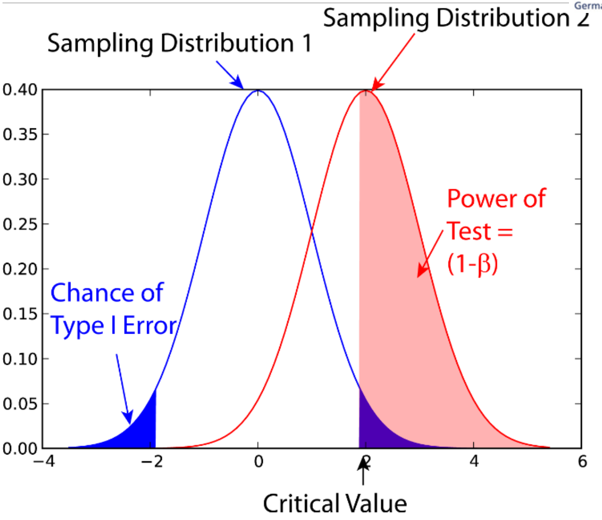


Figure: Illustration of the null distribution of a test statistic.

Copula Goodness-of-fit test

- The empirical (bivariate) copula is defined as the discrete function C_n given by
$$C_n\left(\frac{i}{n},\frac{j}{n}\right)=\frac{\#\left\{(x,y)\mid x\leq x_{(i)} \text{ and } y\leq y_{(j)}\right\}}{n}$$
where $x_{(i)}$ and $y_{(j)}, 1\leq i,j\leq n$ denote the order statistics of the sample and $\#$ provides the cardinality of the subsequent set.
- Cramér-von-Mises statistic: $S_X=\sum_{i=1}^{n}\left(C_X(U_i)-C_n(U_i)\right)^2$
- Selection of the set $\{U_1,\ldots,U_n\}$ is crucial: Set of empirical pseudo-observations? Set of points in a particular region of the copula?
- Null hypothesis: Copula C_X fits well to C_n.

Contents

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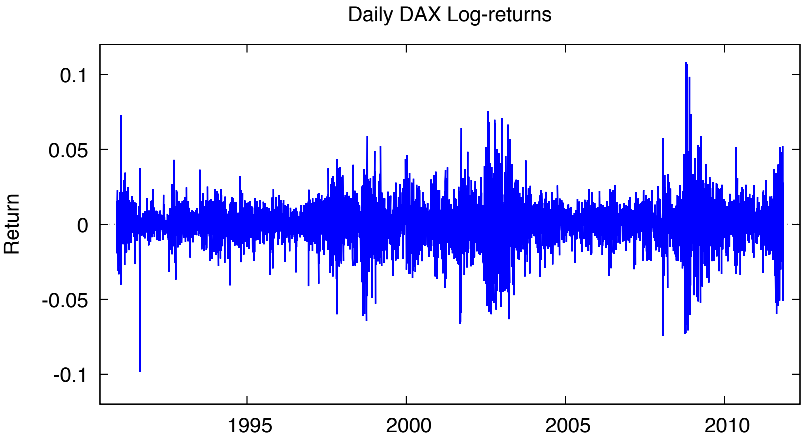
Financial time series

- So far, we have (silently) assumed that a series of log-returns of a risk factor corresponds to an iid sample.
- It turns out that this assumption is not justified for typical financial data.
- **Time series models** are more appropriate for capturing phenomena such as volatility clustering.

Empirical stylised facts

- In a first step, we investigate typical properties, so-called **empirical stylised facts** of financial time series data.
- By a **stylised fact** we mean an empirical observations that applies to the majority of daily series of risk-factors, such as log-returns on equities, indexes, FX rates and commodity prices.
- Empirical stylized facts:
  - Return series are not iid although they show little serial correlation.
  - Series of absolute or squared returns show profound serial correlation.
  - Conditional expected returns are close to zero.
  - Volatility appears to vary over time.
  - Return series are leptokurtic or heavy-tailed.
  - Extreme returns appear in clusters.

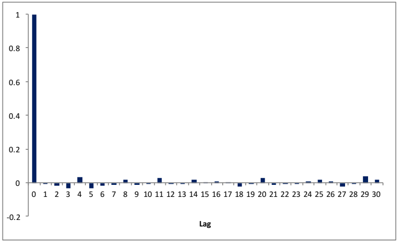
Time-series plot

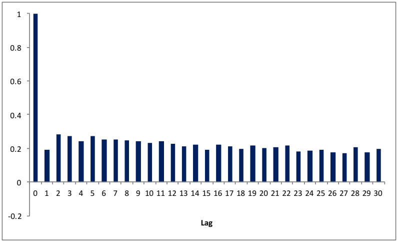


Exercise 5

Discuss the empirical stylised facts apparent in the time-series plot.

- The **autocorrelation** (also **serial correlation**) with lag k of a time series refers to the correlation coefficient of a time series with itself shifted by k time steps.
- The following **correlograms** show the autocorrelations of the DAX returns (left) and the absolute DAX returns (right).





Exercise 6

Discuss the empirical stylised facts apparent in the correlograms.

Empirical stylised facts

- These phenomena become less pronounced as the time period between successive returns is increased.
- It becomes apparent that a time-series model needs to incorporate time-varying volatility, ideally addressing the dependence in the data in an appropriate way.
- We consider three standard approaches to modelling realised volatility:[5]
  - moving averages (MA),
  - exponentially weighted moving average (EWMA) models,
  - generalised autoregressive conditional heteroskedastic (GARCH) models.

Contents

Moving average

- Given daily log returns $x_1, \dots, x_n$, the (annualised) volatility of the sample is defined as
$$\sigma = \sqrt{250} \cdot \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2},$$
where $\bar{x}$ denotes the sample mean.

- The **moving average (MA) process** of historical volatility based on q days is the process of q-day volatility:
$$\sigma_t = \sqrt{250} \cdot \sqrt{\frac{1}{q-1} \sum_{i=1}^{q} (x_{t-i} - \bar{x}_t)^2}, \quad t = q+1, \dots, n,$$
where $\bar{x}_t = \frac{1}{q} \sum_{i=1}^{q} x_{t-i}$ denotes the sample mean based on the previous q days.

Moving average

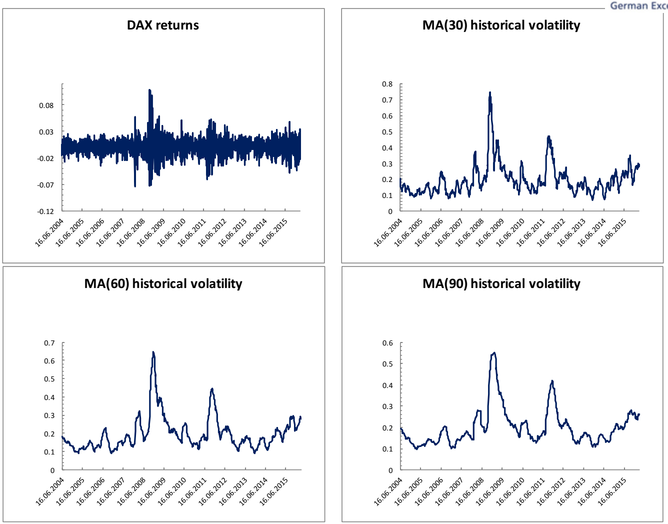


Figure: DAX log-returns and MA volatility processes based on 30, 60 and 90 days.

Contents

Exponentially weighted moving average

- An extension of the moving average model is the **exponentially weighted moving average (EWMA)** model, which gives more weight to recent data and less weight to distant data.
- This approach was proposed by RiskMetrics.
- For observations $x_1, \dots, x_n$ the **EWMA-estimates** for the variance are

$$\hat{\sigma}^2_{\mathrm{EWMA}} = \frac{1-\lambda}{1-\lambda^n} \sum_{i=0}^{n-1} \lambda^i \, (x_{n-i} - \bar{x})^2,$$

  with **decay factor** $0 < \lambda < 1$.
- The decay factor λ controls the speed at which the weights λ^{i-1} decline with increasing i.
- The total weight entering the formula is 1, since $\sum_{i=0}^{n-1} \lambda^i = (1-\lambda^n)/(1-\lambda)$.
- In the limiting case of $\lambda = 1$ we retain in principle the standard estimators, except with the pre-factor $1/n$ in place of $1/(n-1)$.

The n-th partial sum of the geometric series can be calculated as follows:

$$s_n = \sum_{k=0}^{n} \lambda^k = 1 + \lambda + \lambda^2 + \cdots + \lambda^n$$

Multiplying by λ gives:

$$\lambda s_n = \left(\lambda + \lambda^2 + \lambda^3 + \cdots + \lambda^{n+1}\right)$$

Subtracting these two equations gives:

$$s_n - \lambda s_n = \left(1 - \lambda^{n+1}\right)$$

Factoring out s_n:

$$s_n(1-\lambda) = \left(1 - \lambda^{n+1}\right)$$

Proof of $\sum_{i=0}^{n-1} \lambda^i = (1-\lambda^n)/(1-\lambda)$

Dividing by $(1-\lambda)$ for $\lambda \neq 1$ gives the formula we are looking for for the partial sums:

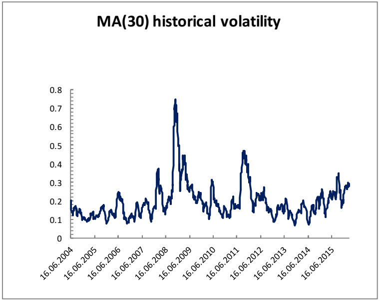
$$s_n = \frac{1-\lambda^{n+1}}{1-\lambda}$$

- Even for many observations $x_1, \dots, x_n$, after a certain index i the values x_{n-i} have very little impact on the result of the EWMA estimate.
- For $i > \log(0.001)/\log(\lambda)$ about 99.9% of the information is already contained in the preceding data.
- Practical experience by RiskMetrics provides the following rules of thumb for λ:

$$\lambda = 0.94 \text{ for estimating a daily variance, covariance}$$
$$\lambda = 0.97 \text{ for estimating a monthly variance, covariance.}$$

- The corresponding relevant lengths of the data history are $112 \approx \log(0.001)/\log(0.94)$ and $227 \approx \log(0.001)/\log(0.97)$.

Exponentially weighted moving average



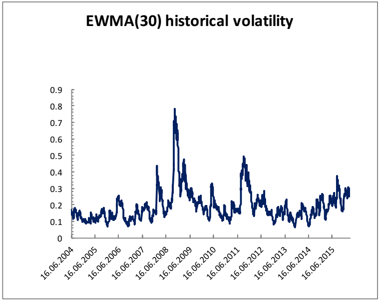


Figure: DAX MA volatility and EWMA volatility (30 days).

Contents

- Aside from volatility clustering, the class of **GARCH (generalised autoregressive conditional heteroskedastic) models** allows for fatter tails than suggested by the normal distribution and incorporates autocorrelation (serial correlation) in absolute/squared returns.
- The **GARCH(1,1) model** is the simplest and most widely used of the family of GARCH-type models.
- A process $X = (X_t)_{t\in\mathbb{Z}}$ is a **GARCH(1,1) process** if it satisfies

$$X_t = \sigma_t Z_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \beta\sigma_{t-1}^2,$$

  where the **innovations** Z_t, $t = 1, 2, \dots$ are iid standard normally distributed, and $\alpha_0 > 0$, $\alpha_1 \geq 0$ and $\beta \geq 0$.
- Note how the volatility σ_t is driven by the current volatility σ_{t-1} and by shocks in the process X_{t-1}^2.

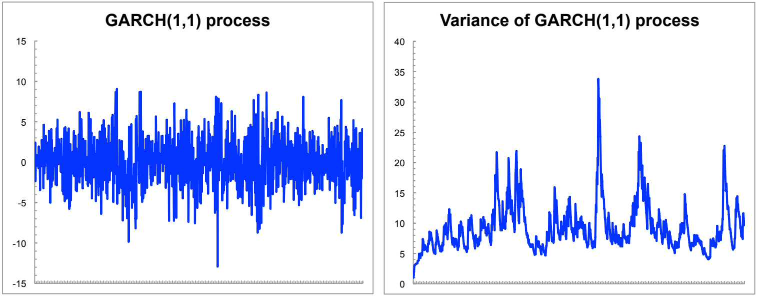


Figure: GARCH(1,1) process with parameters $\alpha_0 = 0.5$, $\alpha_1 = 0.1$, $\beta = 0.85$.

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Theorem 4

Let X be a a GARCH(1,1) process satisfying $\alpha_1 + \beta < 1$. Then, for all $s, t \in \mathbb{Z}$,

(i) $\mathbb{E}(X_t) = 0$;

(ii) $Var(X_t) = \dfrac{\alpha_0}{1 - \alpha_1 - \beta}$ *(unconditional variance);*

(iii) *the* **autocorrelation** $\mathbb{E}(X_t\, X_s)/\sqrt{Var(X_t)\, Var(X_s)}$ *is* 0 *whenever* $s \neq t$;

(iv) *the variance of* X_t *conditional on the information up to time* $t-1$ *is* σ_t^2;

(v) *the kurtosis of* X_t *is*

$$\frac{\mathbb{E}(X_t^4)}{\mathbb{E}(X_t^2)^2} = 3 \cdot \frac{(1 - (\alpha_1 + \beta)^2)}{1 - (\alpha_1 + \beta)^2 - 2\,\alpha_1^2},$$

In particular, X_t *has a positive excess kurtosis.*

Discussion of GARCH properties

- The factor 3 in the kurtosis is just the kurtosis of the standard normal distribution.
- Mean zero is reasonable for daily changes in log-returns. One can add a location parameter μ if wished.
- Constant (unconditional) variance: if we have no information at all, there is no reason to have a time-dependent estimate (why should I assume that Allianz share vol to be higher in five years than in three years?). Proof of property (ii):

$$
\begin{aligned}
\mathrm{Var}(X_t) = \mathbb{E}(X_t^2) = \mathbb{E}(\sigma_t^2 Z_t^2) &\overset{\text{independence}}{=} \mathbb{E}(\sigma_t^2) \cdot 1 \\
&= \alpha_0 + \alpha_1 \mathbb{E}(X_{t-1}^2) + \beta \mathbb{E}(\sigma_{t-1}^2) \\
&= \alpha_0 + (\alpha_1 + \beta)\mathrm{Var}(X_{t-1}).
\end{aligned}
$$

Now write $\mathrm{Var}(X_t)$ as a geometric series. Its limit is known and identical with the results of (ii).

Discussion of GARCH properties

- No autocorrelation is one of the stylised facts.
- Conditional variance σ_t^2: based on information up until $t-1$ we are in a much better position to make an estimate – this is reflected in the time-varying and stochastic variance.
- Positive excess kurtosis means fatter tails.

Important points:

- With just some normally distributed innovations we arrive at a model with fatter tails than the normal distribution!
- With the nice structure involving the normal distribution we can hope to get nice closed formulas for quantities of interest!

The GARCH process and its properties

- The more general GARCH(p,q) model is defined by setting the variance to

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{p} \alpha_i X_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2.$$

- The EWMA model can be viewed as a special GARCH(1,1) process with $\alpha_0 = 0$, $\alpha_1 = 1 - \lambda$ and $\beta = \lambda$.
- There are many extensions of GARCH processes (Integrated GARCH, GARCH with leverage, Threshold GARCH, ...).

Fitting a GARCH model to data

- The standard method for fitting a GARCH model is maximum likelihood.
- Given realisations $x_0, x_1, \dots, x_n$, the joint density of the sample can be written as

$$f(x_1 \dots, x_n | x_0, \sigma_0) = \prod_{t=1}^{n} f_{t|t-1,\dots 0}(x_t | x_{t-1}, \dots, x_0, \sigma_0),$$

  with f the joint density of $x_1, \dots, x_n$ conditional on x_0, σ_0 and $f_{t|t-1,\dots,0}$ the conditional density of x_t given the information up to time $t-1$.
- The conditional densities f_t depend on the past only through the value of σ_t, which is given recursively from $\sigma_0, X_0, \dots, X_{t-1}$ using $\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \beta \sigma_{t-1}^2$.
- This gives the conditional likelihood equation

$$L_{(x_1,\dots,x_n)}(\alpha_0, \alpha_1, \beta) = \prod_{t=1}^{n} \frac{1}{\sigma_t} \mathrm{n}\left(\frac{x_t}{\sigma_t}\right),$$

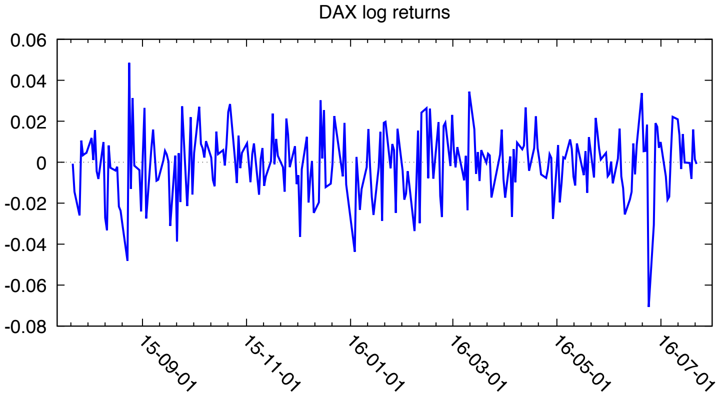
  where $\mathrm{n}(x)$ is the standard normal density and $\sigma_t = \sqrt{\alpha_0 + \alpha_1 X_{t-1}^2 + \beta \sigma_{t-1}^2}$.
- Since the value of σ_0 is not observed one typically chooses as starting value the sample variance of $x_1, \dots, x_n$.

Fitting a GARCH model to data: Variance targeting

- It can happen that the unconditional vola of the modelled GARCH process $\mathrm{Var}(X_t) = \dfrac{\alpha_0}{1-\alpha_1-\beta}$ deviates strongly from the empirically observed one.
- This is discussed in the literature under the term "variance targeting". One possible approach is to restrict α_0 in the parameter estimation such that the modelled and observed variance coincide.
- One other possible approach is to first determine the GARCH parameters without side conditions and afterwards set $\alpha_0 := \mathrm{Var}(X_t) \cdot (1-\alpha_1-\beta)$.
- Some institutions only take this step if the modeled and observed unconditional variance deviate materially.

Example

- We fit daily DAX log-returns of a one-year time period to a GARCH(1,1)-model:



Example

```
Model 1:  GARCH, using observations 2015-07-23-2016-07-22 (T = 255)
Dependent variable:  returns
QML standard errors
```

	Coefficient	Std. Error	z	p-value
const	−0.000417621	0.00102443	−0.4077	0.6835
α_0	0.000120767	7.02454e−05	1.719	0.0856
α_1	0.169563	0.103046	1.646	0.0999
β_1	0.369480	0.297780	1.241	0.2147

Mean dependent var	−0.000498	S.D. dependent var	0.016169
Log-likelihood	694.5721	Akaike criterion	−1379.144
Schwarz criterion	−1361.438	Hannan-Quinn	−1372.022

```
Unconditional error variance = 0.000261993
```

Volatility forecasting

- One use of GARCH models is to forecast future volatility.
- Given returns $x_0, \dots, x_t$, forecasts for the variance are obtained from a GARCH(1,1)-model by (assuming that $\alpha_1 + \beta < 1$):

$$\hat{\sigma}^2_{t+1} = \mathbb{E}(X^2_{t+1} | X_t, \sigma_t) = \hat{\alpha}_0 + \hat{\alpha}_1 X_t^2 + \hat{\beta} \sigma_t^2, \tag{18}$$

  and

$$\hat{\sigma}^2_{t+h} = \mathbb{E}(X^2_{t+h} | X_t, \sigma_t) = \hat{\alpha}_0, \sum_{i=0}^{h-1} (\hat{\alpha}_1 + \hat{\beta})^i + (\hat{\alpha}_1 + \hat{\beta})^{h-1} (\hat{\alpha}_1 X_t^2 + \hat{\beta} \sigma_t^2).$$

- As $h \to \infty$, the variance forecast converges to the unconditional variance $\dfrac{\alpha_0}{1 - \alpha_1 - \beta}$; loosely speaking, the information until time t becomes less relevant as h increases.
- A derivation of this formula is beyond the scope of the course.

Value-at-risk

- Equation (18) can be used to calculate Value-at-risk for the time horizon $\Delta t = [t, t+1]$ for a position with notional amount V_t:

$$\mathbf{VaR}_{\alpha,\Delta t} = -V_t \, \hat{\sigma}_{t+1} \mathrm{N}_{1-\alpha}.$$

- Observe that this approach can be applied to individual positions or to portfolios.
- In case of a portfolio, one generates a time series of the portfolio returns by means of historical simulation and then estimates the corresponding GARCH model.
- Multivariate GARCH models are beyond the scope of this course.

Contents

- It can happen that not only the residuals are return volas are autocorrelated, but also the returns themselves. To address such a property, a GARCH model can be combined with a so-called ARMA model, as discussed below:
- We assume that return time series of interest, Y_t, is decomposed into two parts, the predictable and unpredictable component, $Y_t = \mathbb{E}(Y_t \mid F_{t-1}) + \mathbf{X}_t$, where F_{t-1} is the information set at time $t-1$ and $\mathbb{E}(Y_t \mid F_{t-1})$ is the predictable part. X_t is the unpredictable part, or innovation process.
- The unpredictable component X_t shall be expressed as a GARCH process.
- The modelling of F_{t-1} is usually involved. Our first attempt is the so-called ARMA model:
- The ARMA model (an acronym for Auto-Regressive Moving Average) creates a linear equation which describes and forecasts the time series data. This equation is generated through two separate parts which can be described as:
  - AR (auto-regression): equation terms created based on past data points

- MA (moving average): equation terms of error or noise based on past data points

- An ARMA model is classified as an "ARMA(p, q)" model, where:
  - **p** is the number of autoregressive terms, and
  - **q** is the number of lagged forecast errors in the prediction equation.
  - Formally:

$$Y_t = \underbrace{c + \sum_{i=1}^{p} \varphi_i Y_{t-i} + \sum_{i=1}^{q} \theta_i X_{t-i}}_{=\mathbb{E}\left(Y_t | F_{t-1}\right)} + X_t$$

with $X_{t-i} = \left(Y_{t-i} - \hat{Y}_{t-i}\right)$. X_t is random White Noise.

Introduction to ARMA(p,q) forecasting equation

- ARMA(1,0) = AR(1): first-order autoregressive model
  - The forecasting equation in this case is

$$\hat{Y}_t = c + \varphi_1 Y_{t-1}$$

- ARMA(2,0) = AR(2): second-order autoregressive model
  - In a second-order autoregressive model (ARMA(2,0)), there would be a Y_{t-2} term on the right as well, and so on.

- ARMA(0,1)=MA(1):

– The predicted value of the time series depends on the error made in the previous step:

$$\hat{Y}_t = c + \theta_1 \mathrm{X}_{\mathrm{t}-1} = c + \theta_1 \left(\mathrm{Y}_{\mathrm{t}-1} - \hat{\mathrm{Y}}_{\mathrm{t}-1} \right)$$

Contents

Criteria for model selection I

- Consider a return time series
$$Y_t = \mu_t + X_t, \tag{19}$$
where μ_t is the conditional mean of the process, and X_t is an innovation process with mean zero.
- Suppose again that the innovations are generated as
$$X_t = \sigma_t Z_t, \tag{20}$$
where Z_t is an independent and identically distributed process with mean 0 and variance 1.

Criteria for model selection II

- Consider again as a basic example the ARMA(1,1)-GARCH(1,1) model introduced before:

$$
\begin{aligned}
Y_t &= c + \varphi_1 Y_{t-1} + X_t + \theta_1 X_{t-1} \\
X_t &= \sigma_t Z_t \\
\sigma_t^2 &= \alpha_0 + \alpha_1 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\
Z_t &\sim \text{i.i.d.}(0, 1)
\end{aligned}
$$

- Given the estimated model and the history of the process at time $t-1$, denoted with F_{t-1}, a one-step-ahead estimate/forecast of the conditional mean and variance of Y_t are the following:

$$
\begin{aligned}
\hat{\mu}_{t+1} := \mathbb{E}\left(Y_{t+1} \,|\, F_t\right) &= \hat{c} + \hat{\varphi}_1 Y_t + \hat{\theta}_1 X_t \\
\operatorname{Var}\left(Y_{t+1} \,|\, F_t\right) &= \hat{\alpha}_0 + \hat{\alpha}_1 X_t^2 + \hat{\beta}_1 \hat{\sigma}_t^2
\end{aligned}
$$

- The discussion of model properties requires the following two formal definitions:

Theorem 8.1

Define the residual series as the prediction error, i.e., the difference between the observed return Y_t and the forcast $\widehat{\mu}_t$:

$$\widehat{e}_t := Y_t - \widehat{\mu}_t$$

Note that

$$\widehat{e}_t = Y_t - \widehat{\mu}_t = \mu_t + X_t - \widehat{\mu}_t = X_t + (\mu_t - \widehat{\mu}_t)$$

and that this expression simplifies to $\widehat{e}_t = X_t$, if μ_t is assumed to be zero.

Criteria for model selection IV

Theorem 8.2

Define the standardised residual series. For that purpose, remember that $X_t = \sigma_t Z_t$ and $\widehat{e}_t = X_t + (\mu_t - \widehat{\mu}_t)$. Both imply that

$$\widehat{Z}_t = \frac{X_t}{\widehat{\sigma}_t} = \frac{\widehat{e}_t - (\mu_t - \widehat{\mu}_t)}{\widehat{\sigma}_t}.$$

We denote the right-hand side as "standardised residual" and use this quantity to compute model-implied innovations. Again note that this expression simplifies to $\widehat{Z}_t = \frac{\widehat{e}_t}{\widehat{\sigma}_t}$, if μ_t is assumed to be zero.

- How exactly can we compute the residual series $\widehat{e}_t$ and the series of standardised residuals $\frac{\widehat{e}_t - (\mu_t - \widehat{\mu}_t)}{\widehat{\sigma}_t}$? This happens iteratively!
  - We have:
    - the time series of the observed returns Y_t

Criteria for model selection V

    - the time series of the independent innovations Z_t, which we sampled
    - (some) initial values $\widehat{\sigma}_0$ and Y_0
    - parameter estimates $\widehat{\varphi}_1$, $\widehat{\theta}_1$ of the ARMA model and $\widehat{\alpha}_0, \widehat{\alpha}_1, \widehat{\beta}_1$ of the GARCH model
  - Then, for $t = 1$, compute
    - $\widehat{\sigma}_1$ (using $\widehat{\sigma}_1^2 = \widehat{\alpha}_0 + \widehat{\alpha}_1 X_0^2 + \widehat{\beta}_1 \widehat{\sigma}_0^2$, see page 160)
    - X_1 (using $X_1 = \widehat{\sigma}_1 Z_1$, see page 160)
    - μ_1 (using $Y_1 = \mu_1 + X_1$, see equation (19))
    - $\hat{\mu}_1$ (using $\hat{\mu}_1 = \hat{c} + \hat{\varphi}_1 Y_0 + \hat{\theta}_1 X_0$, see page 160)
  - This allows us to finally calculate $\widehat{e}_1 = Y_1 - \hat{\mu}_1$ and $\frac{\widehat{e}_1 - (\mu_1 - \hat{\mu}_1)}{\widehat{\sigma}_1}$.
  - Do the same iteratively for all other t.
- We are now in a position to summarise important features of a suitable time series model that is implied by equations (19) and (20):

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1. The innovations Z_t are assumed to be independent. Evidence in favor of the assumption is that standardized residuals $\widehat{Z}_t, \widehat{Z}_{t-1}, \ldots$ are not autocorrelated. This is because independence implies uncorrelatedness.
2. The innovations Z_t are assumed to be identically distributed. This in particular includes homoscedasticity (homogeneity of variances). Evidence in favor of this assumption is that $\widehat{Z}_t^2, \widehat{Z}_{t-1}^2, \ldots$ are uncorrelated (bearing in mind that $\mathrm{Var}\left(Z_t\right) = \mathbb{E}\left(Z_t^2\right)$).
3. The innovations Z_t are assumed to follow some predefined distribution, such as standard normal or some (normalised) Student-t distribution. The distribution of the standardised residuals shall be in line with this predefined distribution.
4. The model-implied unconditional variance of Y_t should be in line with the empirical variance of Y_t.

Criteria for model selection VII

5. Depending on the concrete time series model, the impact of an unexpected drop in price (bad news) on the volatility forecast is more or less material than the impact of an unexpected increase in price (good news) on the volatility forecast. For example, a GARCH model is not capable of capturing an asymmetric news impact, where an unexpected decrease in price increases predictable volatility more than an unexpected increase in price (good news) of similar magnitude. In contrast to this example, other time series models, such as apARCH, do additionally capture asymmetry in return volatility.

When choosing an appropriate time series model, we do not only compare the BIC value of calibrated alternative models, but also investigate for each model candidate, if the calibrated models fulfill the theoretical model properties 1.-5. Additionally, a time series model can only be soundly used for volatility prediction, if its parameters are stable over time. We therefore additionally test the stability of the model parameters over time:

6. The parameter estimates of the time series model are stable over time.

Criteria for model selection VIII

We now discuss the validation of model properties (1)–(5) using tests, which in parts utilise residuals and standardised residuals:

- Ad 1.: We can use the standardised residuals to investigate their autocorrelation with the Ljung-Box test (G.M. Ljung and G.E.P. Box (1978). On a measure of lack of fit in time series models. Biometrika, 65, 297–303.). The null hypothesis for this test is that the first m autocorrelations of the standardised residuals are jointly zero, $H_0 : \rho_1 = \rho_2 = \cdots = \rho_m = 0$. The alternative hypothesis would contradict the modelling assumption that the z_t are i.i.d. An alternative to the Ljung-Box test, which places equal weight to the different lags investigated, the Weighted Ljung-Box test (T.J. Fisher and C.M Gallagher. New weighted portmanteau statistics for time series goodness of fit testing. Journal of the American Statistical Association, 107(498):777–787, 2012.) puts more emphasis on the first autocorrelation, and the least emphasis on the autocorrelation at lag m.

Criteria for model selection IX

- Ad 2.: We can apply the (Weighted) Ljung-Box test to the squared standardised residuals. As an alternative, we can investigate autocorrelation in the squared standardised residuals using Engle's ARCH test, regressing

$$\widehat{Z}_t^2 = \alpha_0 + \alpha_1 \widehat{Z}_{t-1}^2 + \cdots + \alpha_m \widehat{Z}_{t-m}^2 + u_t,$$

  where u_t is a white noise error process. The null hypothesis is

$$H_0 : \alpha_0 = \alpha_1 = \cdots = \alpha_m = 0$$

- Ad 3.: After model calibration, the distribution of the standardised residuals shall be in line with the one assumed during model calibration, such as $Z_t \sim \mathcal{N}(0, 1)$. This can be tested with a goodness-of-fit test, such as Kolmogorov-Smirnov or the Adjusted Pearson goodness-of-fit test (F.C. Palm. Garch models of volatility. Handbook of Statistics, 14:209–240, 1996.).

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- Ad 4.: This question does not require a particular test. The consequence of a material mismatch between modelled and observed unconditional variance could be variance targeting, with which the intercept in the conditional variance model (e.g. α_0 in the GARCH(1,1) example above) is adjusted accordingly to align the modelled and observed unconditional variances.
- Ad 5.: The symmetry of the news impact can be tested with Engle's sign bias test (R.F. Engle and V.K. Ng. Measuring and testing the impact of news on volatility. Journal of Finance, 48(5):1749–1778, 1993.), which regresses the squared standardised residuals on lagged negative and positive shocks as follows:

$$\hat{Z}_t^2 = c_0 + c_1 I_{\hat{e}_{t-1}<0} + c_2 I_{\hat{e}_{t-1}<0}\hat{e}_{t-1} + c_3 I_{\hat{e}_{t-1}\geq 0}\hat{e}_{t-1} + u_t,$$

where I is the indicator function. The null hypotheses are $H_0 : c_i = 0$ (for $i = 1$: Sign Bias test focusing on the impact of **positive and negative** return shocks on volatility not predicted by the model under consideration, $i = 2$:

Criteria for model selection XI

Negative Sign Bias test focusing on the effects that large and small **negative** return shocks have on volatility which is not predicted by the volatility model, $i = 3$: Positive Sign Bias test focussing on the different impacts that large and small **positive** return shocks may have on volatility, which are not explained by the volatility model), and that jointly $H_0 : c_1 = c_2 = c_3 = 0$.

- Ad 6.: The stability of the model parameters over time can be tested with the Nyblom stability test (J. Nyblom. Testing for the constancy of parameters over time. Journal of the American Statistical Association, 84(405):223–230, 1989.) that uses the null hypothesis that all parameters of the time series model are stable over time.

All these tests are e.g. implemented in the R-package "rugarch".

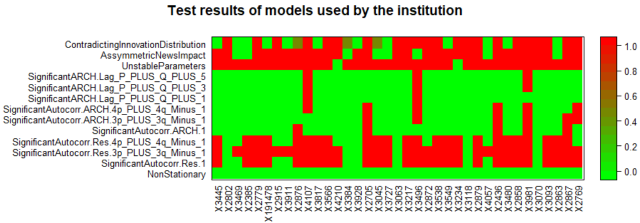

Figure: Test results of funds time series: Model used by the inspected institution

Criteria for model selection XIII

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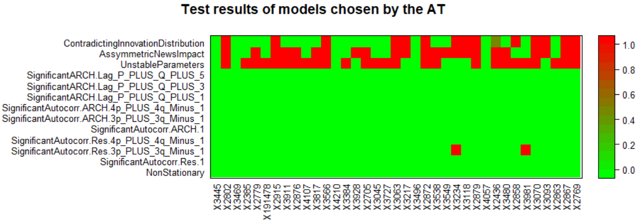


Figure: Test results of funds time series: Model used by the assessment team

Contents

- The results below are obtained from the R code "Citi.R" shared on Canvas. This code was sourced from "https://www.kaggle.com/ionaskel/value-at-risk-estimation-using-garch-model/report"

Case study: Value at Risk of Citi stock II

- Step 1: Illustrate the data

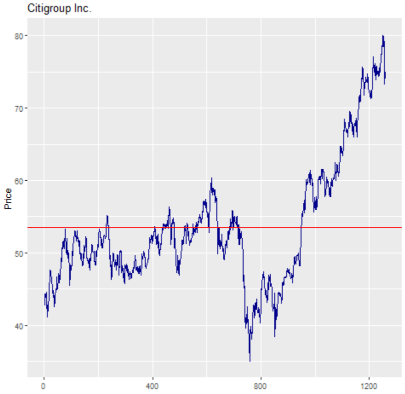


Figure: Time series of Citi stock price.

- Step 2: Compute returns
- Step 3: Have R automatically determine the optimal p and q. This is technically achieved by AIC comparison. Result: ARMA$(2, 0)$ with zero mean is identified to be the best model. R estimates the parameters of the ARMA$(2, 0)$-model with maximum likelihood and obtains that:

$$Y_t = 0.0437 \cdot Y_{t-1} - 0.0542 \cdot Y_{t-2} + X_t.$$

  where X_t is White Noise.
- Step 4: Test the residuals X_t for (an absence of) autocorrelation using the Ljung-Box test. Result: We cannot show that the serial autocorrelation is significant.

- Step 5: Model the noise with GARCH(1,1), assuming that

$$X_t = \sigma_t \cdot Z_t$$

where Z_t is a sequence of independently and identically distributed random variables with zero mean and variance equal to 1 and

$$\sigma_t^2 = \alpha_0 + \alpha_1 \cdot X_{t-1}^2 + \beta_1 \cdot \sigma_{t-1}^2.$$

With successive replacement of the σ_{t-1}^2 term, the GARCH equation can be written as:

$$\sigma_t^2 = \frac{\omega}{1-\beta_1} + \alpha_1 \cdot \sum_{i=1}^{\infty} \beta_1^{i-1} \cdot X_{t-i}^2$$

When we replace with the coefficient estimates given by optimization in R we get the following equation:

$$\sigma_t^2 = 0.000087 + 0.108 \cdot \left(X_{t-1}^2 + 0.825 \cdot X_{t-2}^2 + 0.680 \cdot X_{t-3}^2 + 0.561 \cdot \mathbf{X}_{t-4}^2 + \dots \right).$$

- Step 6: VaR estimation

For the estimation of VaR we use the conditional variance given by GARCH $(1, 1)$ model and determine the VaR using

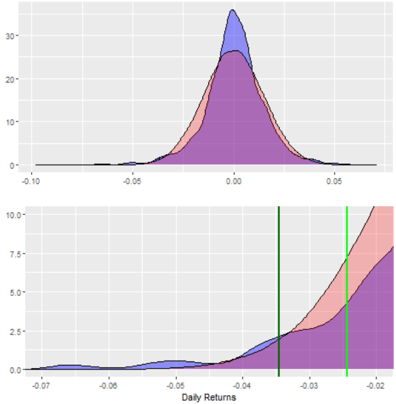
$$VaR(\alpha) = \mu + \hat{\sigma}_{t|t-1} \cdot F^{-1}(\alpha),$$

where $\hat{\sigma}_{t|t-1}$ is the conditional standard deviation given the information at $t-1$ and F^{-1} is the inverse PDF function of the return distribution.

Is the return distribution normal? No, it is not, as illustrated in the figure below, which shows density plots for stock returns (blue) and normal distributed data (red). Vertical lines of the lower plot represent the normal corresponding quantile for $\alpha = 0.05$ (light green) and $\alpha = 0.01$ (dark green). The lower plot indicates that for 95% significance, the usage of a normal distribution overestimates the Value-at-Risk (5% normal quantile: -2.4772%, i.e.m, 95%-VaR=2.477. Empirical 5% quantile: -2.3578%, i.e., 95%-VaR=2.3578). The reason is that the empirical density has less probability mass on the left-hand side of the light green line than the normal distribution. Thus, the light-green line must correspond to a an quantile of the empirical distribution lower than 5%.

However, for a 99%-confidence level, a normal distribution would underestimate the VaR (1% normal quantile: -3.5035%, i.e., 99%-VaR=3.5035. Empirical 1% quantile: -3.9904%, i.e., 99%-VaR=3.9904%). The reason is that the empirical density has more probability mass on the left-hand side of the dark green line than the normal distribution. Indeed: The 1%-quantile of the normal distribution roughly corresponds to a 2% quantile of the empicial distribution.

Case study: Value at Risk of Citi stock VII



Therefore, given that we compute risk on a 95% confidence level here and that we observed above that the normal distribution tends to overestimate risk on a 95% confidence level, let us estimate the quantile of an appropriate Student-t distribution with a standard deviation of 1 and mean equal to 0. This can be done in the rugarch package in R, which estimates that the 95%-quantile of the such a Student-t distribution amounts to -1.485151 instead of the standard normal quantile of -1.644854. This fitted Student-t distribution has 3.73 degrees of freedom.

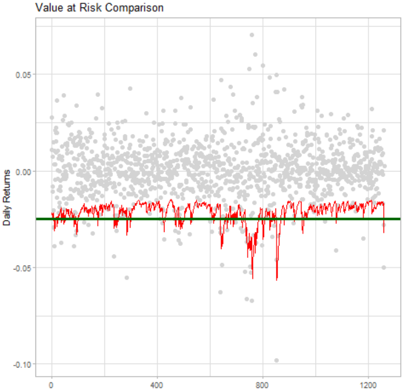


Figure: Time series of the VaR. Red: ARMA+GARCH. Green: Normal VaR.

- Step 7: Rolling estimation of the GARCH parameters based on the 500 most recent observations and rolling moving 1-step ahead forecast of the conditional standard deviation $\hat{\sigma}_{t+1|t}$. We re-estimate GARCH parameters every 50 observations.

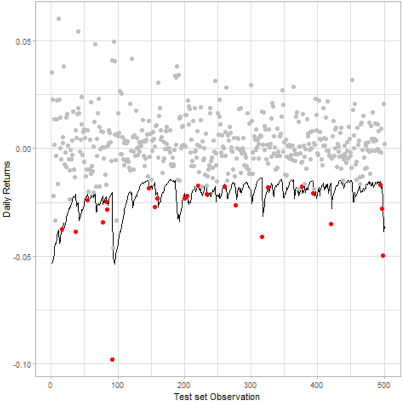


Figure: Time series of the VaR based on rolling parameter estimation.

- Step 8: Backtesting

  Let N denote the observed number of exceptions in the sample. The number of VaR breaches can be shown to follow a binomial distribution, $B(T, p)$. The expected number is 25 (=500 obs. x 5%). Two red lines in the next graph denote the 95% confidence level, the lower being 16 and the upper 35. Therefore, when we check the exceptions on the test set, we expect a number between 16 and 35 to consider the GARCH model to be successfully predictive.

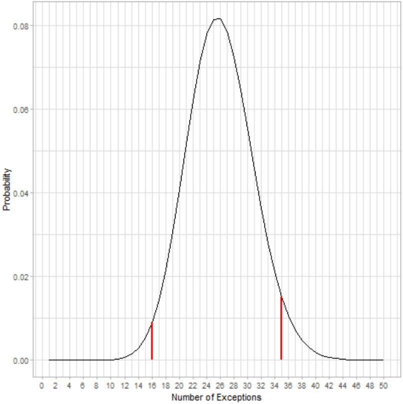


Figure: Distribution of expected VaR-overshootings

Number of VaR-overshootings of the normal VaR: 14

Number of VaR-overshootings with ARMA+GARCH approach: 23

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Exercise 7

Is the "square-root-of-time" scaling rule for volatility justified when volatility is time-varying? Discuss!

Exercise 8

Take a time series of your favourite portfolio and calculate **VaR** at $\alpha = 95\%$ and $\alpha = 99\%$ using the following approaches:

- assuming t-distributed returns
- Historical simulation
- MC simulation
- GARCH(1,1)
- ARMA + GARCH(1,1)

Compare the results.

Contents

Overview

Consider the following problems:

- What is the appropriate height for a sea dyke in Holland?
- What is the expected height of a "freak wave"?
- What is the ultimate 100m world record?
- How can a re-insurer calculate insurance premia to sufficiently cover catastrophic events?
- How can we appropriately measure (extremal) risk in a financial portfolio?
- How can we protect against extremal events in asset management?

- How can we deal – at least statistically – with events that we have no experience with?
- If we don't have data..., we need theory.
- **Extreme Value Theory (EVT)** is concerned with measuring and modelling extreme events or extreme risks.
- There are two main kinds of models for extreme values:
  - the **block maxima models**, which are models for the largest observations collected from large samples of iid observations; applications in risk management include **stress losses**;
  - the **threshold exceedance models**, which are models for all large observations that exceed some high level; applications in risk management include value-at-risk and expected shortfall estimation.

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Matthews, R. (1996) Far out forecasting. In: The New Scientist 2051, October 12, 1996, pp. 36–40

> There is always going to be an element of doubt, as one is extrapolating into areas one doesn't know about. But what EVT is doing is making the best use of whatever data you have about extreme phenomena. (Richard Smith)

> The key message is that EVT cannot do magic, but it can do a whole lot better than empricial curve fitting and guesswork. My answer to the sceptics is that if people aren't given well-founded methods like EVT, they'll just use dubious ones instead. (Jonathan Tawn)

Contents

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Maxima

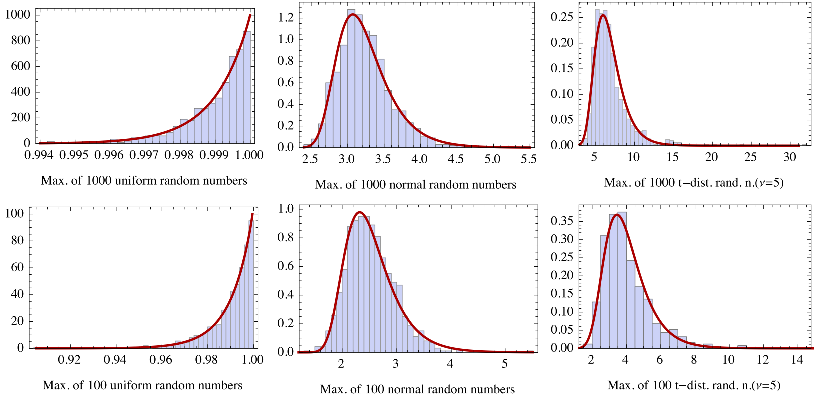
- Let $X_1, X_2, \dots, X_n$ be a sequence of iid random variables with mean μ and variance $\sigma^2 < \infty$.
- These variable could for instance represent financial losses.
- Before we turn to analysing the maximum, recall the **Central Limit Theorem (CLT)**, which states that the **sums** $S_n = X_1 + \cdots + X_n$, $n > 0$, when properly scaled and shifted converge to a normal distribution:

$$\lim_{n\to\infty} \mathbf{P}\left(\frac{S_n - n\,\mu}{\sqrt{n}\sigma} \leq x\right) = \mathrm{N}(x), \quad x \in \mathbb{R}.$$

- Much of the importance of the normal distribution in probability theory comes from its role in the CLT.
- Classical EVT is concerned with limiting distributions for scaled and shifted **maxima** $M_n = \max(X_1, \dots, X_n)$, sometimes also referred to as **block maxima**.

Maxima

- The figure below shows the outcomes of a simulation experiment with block maxima from different distributions.



Max. of 1000 uniform random numbers
Max. of 1000 normal random numbers
Max. of 1000 t−dist. rand. n.(ν=5)
Max. of 100 uniform random numbers
Max. of 100 normal random numbers
Max. of 100 t−dist. rand. n.(ν=5)

Exercise 9

Can you find explanations for the distribution shapes of the maxima?

GEV distribution

- In the case of uniform random numbers on the $[0, 1]$ interval, the maximum turns out to be close to 1.
- Generally, the distribution of the maximum is close to the $1/n$-quantile, where n refers to the block size.
- It turns out that, for a large class of distributions,[6] the maximum converges to one out of only three distributions.
- These three distributions are comprised in the so-called **generalised extreme value (GEV) distribution**.
- In other words, for maxima the GEV distribution plays a role similar to the normal distribution in case of sums.

[6] Virtually all distributions that we commonly work with in finance.

Definition 5

The distribution function of the standard GEV distribution is given by

$$H_\xi(x) = \begin{cases} \mathbf{e}^{-(1+\xi x)^{-1/\xi}}, & \xi \neq 0, \\ \mathbf{e}^{-e^{-x}}, & \xi = 0, \end{cases}$$

where $1 + \xi x > 0$. A three-parameter family is obtained by defining $H_{\xi,\mu,\sigma}(x) := H_\xi((x-\mu)/\sigma)$ for a location parameter $\mu \in \mathbb{R}$ and a scale parameter $\sigma > 0$.

A three-parameter family is obtained by defining

$$H_{\xi,\mu,\sigma}(x) := H_\xi\left(\frac{x-\mu}{\sigma}\right)$$

for a location parameter $\mu \in \mathbb{R}$ and a scale parameter $\sigma > 0$.

GEV distribution

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- ξ: **shape parameter** that defines type of distribution:

Fréchet distribution	$\xi > 0$	"heavy-tailed"
Gumbel distribution	$\xi = 0$	"light-tailed"
Weibull distribution	$\xi < 0$	"short-tailed"

GEV distribution

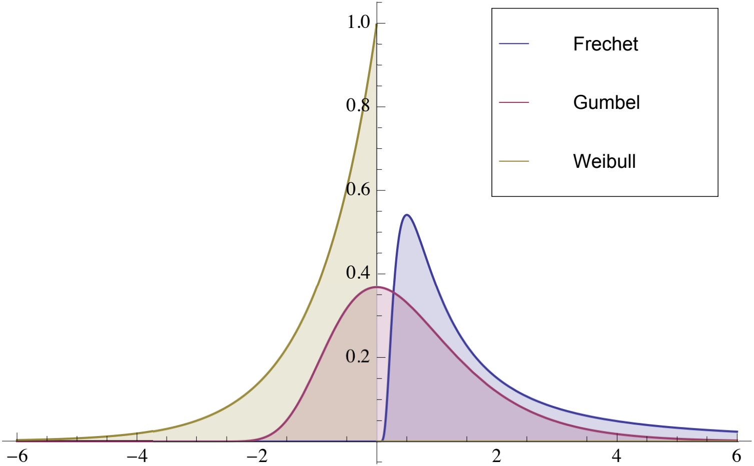

Figure: The three types of GEV distributions: The Fréchet distribution $(\xi > 0)$, the Gumbel distribution $(\xi = 0)$ and the Weibull distribution $(\xi < 0)$.

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- The **Fisher-Tippett Theorem** makes our findings precise:

Theorem 6 (Fisher-Tippett Theorem)

Let (X_n) *be a sequence of iid random variables, and let* $M_n = \max(X_1, \dots, X_n)$. *If there exist norming constants* $c_n > 0$, $d_n \in \mathbb{R}$ *and some non-degenerate distribution function* H *such that*

$$\lim_{n\to\infty} \mathbf{P}\left(\frac{M_n - d_n}{c_n} \leq x\right) = H(x), \tag{21}$$

then H *is a distribution of type* H_ξ, *that is a GEV distribution.*

Max-domain (MDA)

- The distribution function F of the X_i is said to be in the **maximum domain of attraction (MDA)** of H.
- Examples of distributions in the respective MDA's are:
  - MDA of the Fréchet distribution: Pareto, Student t, loggamma, Cauchy distributions;
  - MDA of the Gumbel distribution: normal, lognormal, gamma distributions;
  - MDA of the Weibull distribution: uniform, beta distributions.
- The theory can be generalised to certain types of non-iid time series including GARCH processes.

Fitting the GEV distribution

- Suppose we have (a lot of) data sampled independently from a distribution F, which we assume lies in the MDA of an GEV distribution H_ξ.
- The **block maxima method** for data to fitting $H_{\xi,\mu,\sigma}$ partitions the data into consecutive blocks of size n, say m blocks, and chooses the maxima of the blocks, $M_{n1}, M_{n2}, \dots, M_{nm}$ as the data sample.
- The Fisher-Tippett Theorem suggests that, provided n is large enough, the distribution of the n-block maxima $M_{n1}, \dots, M_{nm}$, can be approximated by a three-parameter GEV distribution $H_{\xi,\mu,\sigma}$.
- The GEV distribution can be fitted using e.g. maximum likelihood estimation.
- If n is large, then we can assume even for dependent data, that the block maxima are independent.

- Writing $h_{\xi,\mu,\sigma}$ for the density of the GEV distribution, the log-likelihood function is

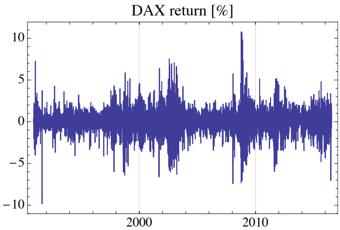
$$\ell_{(M_{n1},\ldots,M_{nm})}(\xi,\mu,\sigma) = \sum_{i=1}^{m} \ln h_{\xi,\mu,\sigma}(M_{ni})$$

$$= -m\ln\sigma - \left(1+\frac{1}{\xi}\right)\sum_{i=1}^{m}\ln\left(1+\xi\frac{M_{ni}-\mu}{\sigma}\right) - \sum_{i=1}^{m}\left(1+\xi\frac{M_{ni}-\mu}{\sigma}\right)^{-1/\xi},$$

with the parameter constraints $\sigma > 0$ and $(1+\xi(M_{ni}-\mu)/\sigma) > 0$, for all i.[7]

[7] Although the parameter space depends on the sample data, consistency holds when $\xi > -1/2$.

Example



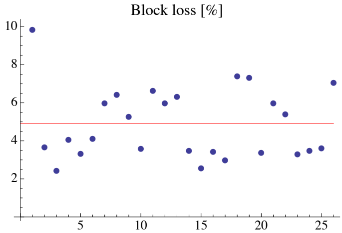


Figure: Left: Daily DAX log-returns. Right: Annual maxima of daily falls in index; red line: mean of annual maxima.

Example

- Using annual block maxima of DAX log-returns starting from 1990, the 26 observations of annual maximal losses were fitted to a GEV distribution,[8] yielding parameters $\hat{\hat{\xi}} = 0.164714$, $\hat{\sigma} = 1.27626$ and $\hat{\mu} = 3.92497$.
- This means that the fitted distribution is a heavy-tailed Fréchet distribution, with infinite seventh moment (the last property follows from $1/\xi = 6.0711 < 7$).

[8] To get a more stable estimate, the location parameter μ was chosen such that the expectation matches the sample mean via the relationship:

$$\mathbb{E}M_n = \mu + \frac{\sigma}{\xi}\left(\Gamma(1-\xi) - 1\right),$$

which denotes the expectation of the GEV for $\xi < 1$ and $\xi \neq 0$ (here, Γ is the Gamma function). The remaining parameters ξ and σ were estimated via Maximum Likelihood.

Stress losses

- The fitted model can be used to analyse **stress losses** in two ways:
- **Return level:** Given a frequency of occurrence of the stress event, estimate the magnitude; more precisely, the k n-block return level $r_{n,k}$ is the $(1 - 1/k)$-quantile of the distribution of the n-block maximum. Roughly speaking the k n-block return level can be interpreted as the level that is exceeded in one out of every k n-blocks on average.
- **Return period:** Given the size of the stress event, estimate the frequency of its occurrence; more precisely, the return period of the event $\{M_n > u\}$ is given by $k_{n,u} = 1/(1 - H(u))$. This implies that the $k_{n,u}$ n-block return level is u.

Example

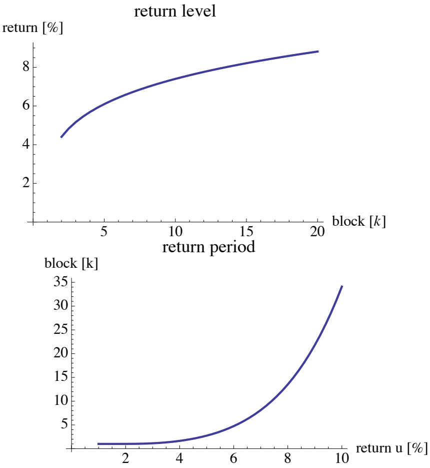


Figure: Top: k annual block return level $r_{k,250}$. Bottom: return period $k_{n,u}$.

Example

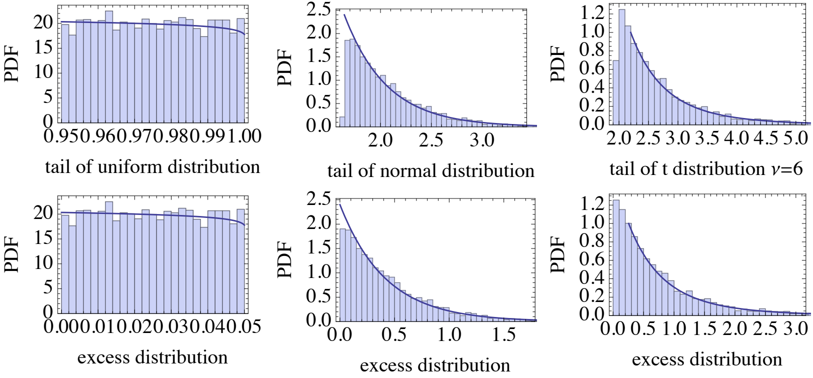
- In the DAX example, the 10-year return level is 7.40%, and the return period for a loss of 5% is 2.74 years. The return levels and return periods are shown in the Figure on the previous page.

Contents

Threshold exceedances

- A disadvantage of the block maxima method is its wasteful use of data.
- The method of **threshold exceedances** uses all data that are extreme in the sense that they exceed a particular designated high level.

- The figure shows the outcomes of a simulation experiment where only the values exceeding the 0.95-quantile are considered. The figure shows both the outcomes (top) and the exceedances beyond a threshold (bottom) of $u = 0.95$.



- Again a limit theorem argument helps to select a model; this time the distribution in question is the **generalised Pareto distribution (GPD)**.

Definition 7

The distribution function of the GPD is given by

$$G_{\xi,\beta}(x) = \begin{cases} 1-(1+\xi x/\beta)^{-1/\xi}, & \xi \neq 0, \\ 1-\mathrm{e}^{-x/\beta}, & \xi = 0, \end{cases}$$

where $\beta > 0$, $x \geq 0$ when $\xi \geq 0$ and $0 \leq x \leq -\beta/\xi$ when $\xi < 0$.

- The parameters ξ and β are the **shape** and the **scale** parameters.

The GPD distribution

- The GPD comprises the following distributions:
  - $\xi > 0$: $G_{\xi,\beta}$ is a Pareto distribution with parameters $1/\xi$ and β/ξ;
  - $\xi = 0$: $G_{\xi,\beta}$ is an exponential distribution;
  - $\xi < 0$: $G_{\xi,\beta}$ is a Pareto type II distribution (which is short-tailed).
- $G_{\xi,\beta}$ is in the MDA of H_ξ, $\xi \in \mathbb{R}$, where H_ξ is a GEV distribution.
- If we model the tail of a distribution with a GPD, this implies the parameter ξ has the same interpretation for both the block maxima and the threshold exceedances methods.

The GPD distribution

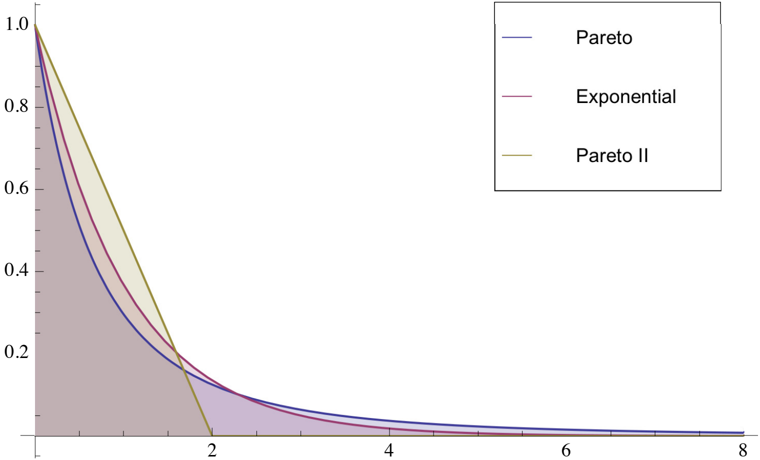


Figure: The three types of GPD distributions: Pareto ($\xi = 0.5$), exponential ($\xi = 0$) and Pareto type II ($\xi = -0.5$); $\beta = 1$.

- The GPD serves as a model for the **excess distribution** over a high threshold.

Definition 8

- For a random variable X, resp. distribution function F, the **excess distribution** $F_u(x)$ over a threshold u is defined as

$$F_u(x) = \mathbf{P}(X - u \leq x | X > u) = \frac{F(x+u) - F(u)}{1 - F(u)},$$

with $0 \leq x \leq x_F - u$ where x_F the right-endpoint of F.[9]

- For a random variable X with finite mean, the **mean excess function** is defined as

$$e(u) = E(X - u | X > u).$$

[9] For unbounded distributions the right-endpoint is ∞.

- If X follows a GPD $F = G_{\xi,\beta}$:

$$F_u(x) = G_{\xi,\beta(u)}(x), \quad \beta(u) = \beta + \xi u, \tag{22}$$

where $0 \leq x < \infty$ if $\xi \geq 0$ and $0 \leq x \leq -(\beta/\xi) - u$ if $\xi < 0$.

- In particular, the **shape parameter** ξ **is constant** for the excess distribution of a GPD and the **scaling parameter** $\beta(u)$ **grows linearly** with the threshold u.
- The mean excess function of the GPD is easily computed from its mean $\mathbb{E}(X) = \beta/(1-\xi)$ and excess distribution to be

$$e(u) = \frac{\beta + \xi u}{1 - \xi}, \tag{23}$$

where $0 \leq u < \infty$ if $0 \leq \xi < 1$ and $0 \leq u \leq -\beta/\xi$ if $\xi < 0$.

- The mean excess function is **linear** in u, which also characterises the GPD.

The following result explains the importance of the GPD:

Theorem 9 (Pickands-Balkema-de Haan)

There exists a positive function $\beta(u)$ such that

$$\lim_{u \to x_F} \sup_{0 \leq x < x_F - u} |F_u(x) - G_{\xi,\beta(u)}(x)| = 0,$$

if and only if F is in the MDA of H_ξ, $\xi \in \mathbb{R}$, with x_F the right-endpoint of F.

- In the general case, the functional form of $\beta(u)$ is not known.
- However, if $\beta(u)$ is linear, for u sufficiently large, then the GPD serves as a good model for the tail of the distribution.

Fitting the GPD distribution

- We make the Pickands-Balkema-de Haan Theorem by asserting that for some fixed threshold u that the GPD serves as a model for the excess distribution of the data, that is, $F_u(x) = G_{\xi,\beta}(x)$.
- For the moment we take u as fixed.
- Given iid loss data $x_1, \dots, x_n$, a (random) number N_u will exceed u; relabel these as $\tilde{x}_1, \dots, \tilde{x}_{N_u}$ and set the excess loss to $y_j = \tilde{x}_j - u$.
- Writing $g_{\xi,\beta}$ for the density of the GPD, the log-likelihood function is given by

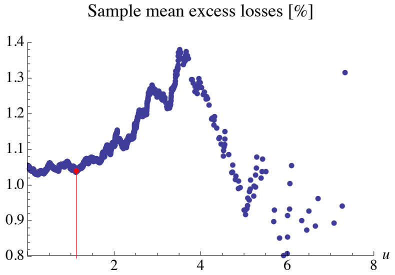
$$\ell_{(y_1,\dots,y_{N_u})}(\xi,\beta) = \sum_{j=1}^{N_u} \ln g_{\xi,\beta}(y_j)$$

$$= -N_u \ln \beta - \left(1 + \frac{1}{\xi}\right) \sum_{j=1}^{N_u} \ln\left(1 + \xi \frac{y_j}{\beta}\right),$$

to be maximised subject to the constraints $\beta > 0$ and $1 + \xi Y_j \beta > 0$ for all j.

Example

- Consider the DAX data used in the block maxima approach (Figure 22).
- Assuming that for a threshold large enough the tail of the loss data follows a GPD distribution, the mean excess function will be linear in u.
- We determine the threshold u from the mean excess function of the data:



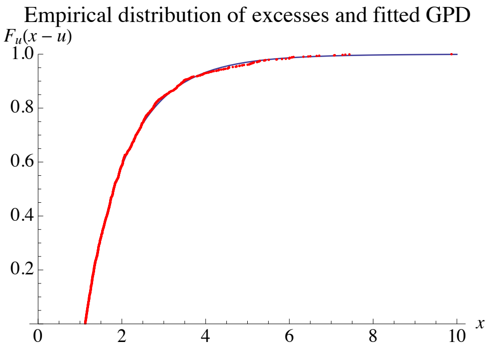
- For large thresholds ($u > 4$), there are too few data points to warrant a proper estimation.
- From the smaller values of u we pick the one that is leftmost of what appears to be a line. This gives $u = 1.125$.

Example

- The visual identification of an appropriate u has a major drawback: It is hardly possible to automate this approach.
- This is a particular issue in a simulation study that computes a daily VaRs over a period of several years.
- As a potential workaround: Define u as the $(1-\alpha)$-quantile of the P&L-distribution and validate the VaR-impact of alternative selections of α. If we are lucky enough, the choice of u (i.e., the choice of the confidence level α) - within sensible bounds, of course - is not crucial.

Example

- The MLE estimates are $\hat{\xi} = 0.0732$ and $\beta = 0.96$.
- The fitted GPD distribution is shown in the following figure:



- The GPD approach can be used to calculate value-at-risk and expected shortfall in close form.
- The **tail probabilities** of a GPD F are given by

$$1 - F(x) = (1 - F(u)) \cdot \left(1 + \xi \frac{x-u}{\beta}\right)^{-1/\xi}, \quad x \geq u.$$

- For $\alpha \geq F(u)$ we can back out $\mathbf{VaR}_\alpha$ to be

$$\mathbf{VaR}_\alpha = q_\alpha(F) = u + \frac{\beta}{\xi}\left(\left(\frac{1-\alpha}{1-F(u)}\right)^{-\xi} - 1\right) \approx u + \frac{\beta}{\xi}\left(\left(\frac{1-\alpha}{N_u/N}\right)^{-\xi} - 1\right),$$

and for $\xi < 1$ expected shortfall can be calculated as

$$\mathbf{ES}_\alpha = \frac{1}{1-\alpha}\int_\alpha^1 q_x(F)\,\mathrm{d}x = \frac{\mathbf{VaR}_\alpha}{1-\xi} + \frac{\beta - \xi u}{1-\xi}.$$

- We know already that the following holds:
  1. $F_u(x) = \mathbf{P}(X - u \le x | X > u) = \frac{F(x+u) - F(u)}{1 - F(u)} = \frac{\mathbf{P}(X \le x + u \cap X > u)}{\mathbf{P}(X > u)}$
  2. $F_u(x) \approx G_{\xi,\beta(u)}(x)$ according to Pickands-Balkema-de Haan
  3. $F(u) = \mathbf{P}(X \le u) \approx \frac{N - N_u}{N}$, where N denotes the total number of data and N_u the number of exceedances over threshold u.
- Combining 1., 2. and 3. yields

$$\begin{aligned} F(x+u) &= (1 - F(u))F_u(x) + F(u) \\ &\approx (1 - F(u))G_{\xi,\beta(u)}(x) + F(u) \\ &\approx \frac{N_u}{N} G_{\xi,\beta(u)}(x) + \frac{N - N_u}{N} \end{aligned} \tag{24}$$

Proof of the VaR formula II

- It per definition of the *VaR* holds that

$$\mathbf{P}(X \leq VaR_\alpha) = F(VaR_\alpha) = \alpha.$$

- Let x_p such that $x_p + u = VaR_\alpha$. Then,

$$\begin{aligned}
\alpha = F(x_p + u) &\approx \frac{N_u}{N} G_{\xi,\beta(u)}(x_p) + \frac{N - N_u}{N} \\
\Leftrightarrow \alpha - \frac{N - N_u}{N} &\approx \frac{N_u}{N} G_{\xi,\beta(u)}(x_p) \\
\Leftrightarrow \alpha - 1 + \frac{N_u}{N} &\approx \frac{N_u}{N} G_{\xi,\beta(u)}(x_p)
\end{aligned}$$

Proof of the VaR formula III

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$$\Leftrightarrow \frac{(\alpha-1)N}{N_u} + \frac{N_u \cdot N}{N \cdot N_u} \approx G_{\xi,\beta(u)}(x_p) = 1 - (1+\xi x_p/\beta)^{-1/\xi}$$

$$\Leftrightarrow \frac{(\alpha-1)N}{N_u} \approx -(1+\xi x_p/\beta)^{-1/\xi}$$

$$\Leftrightarrow \frac{(1-\alpha)N}{N_u} \approx (1+\xi x_p/\beta)^{-1/\xi}$$

$$\Leftrightarrow \left(\frac{(1-\alpha)N}{N_u}\right)^{-\xi} \approx 1+\xi x_p/\beta$$

$$\Leftrightarrow \left(\left(\frac{(1-\alpha)N}{N_u}\right)^{-\xi} - 1\right)\frac{\beta}{\xi} \approx x_p$$

Example

- For the example of the DAX data, we obtain

$$\mathbf{VaR}_{99\%} = 4.10\% \quad \text{and} \quad \mathbf{ES}_{99\%} = 5.37\%.$$

- To translate this into a capital requirement, we assume for simplicity that the returns are discrete, so that $\Delta V = V_{\Delta t} - V_0 = V_0\, r$, and we obtain (setting $r = -\mathbf{VaR}_{99\%}$ and $r = -\mathbf{ES}_{99\%}$, respectively, and with $V_0 = 10147.5$)

$$\mathbf{VaR}_{99\%}(-\Delta V) = 415.81 \quad \text{and} \quad \mathbf{ES}_{99\%}(-\Delta V) = 545.14.$$

- For comparison, we calculate VaR and ES in some other models:

	$\mathbf{VaR}_{99\%}$	$\mathbf{ES}_{99\%}$
Delta-Normal approach	339.04	388.43
Historical simulation	433.14	552.02
GARCH	348.15	398.86

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